

Urban structure, housing stock and gentrification dynamics in the Brest metropolitan area:

does the distribution of the social housing stock reinforce or mitigate residential segregation?

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ABSTRACT

Using the "Individus localisés au canton-ou-ville" file from the 2022 INSEE population census, this work analyses the effect of the distribution of the social housing stock on residential segregation in the Brest metropolitan area. The analysis rests on a two-scenario comparison, without and then with the residents of the social stock, applied to three socio-occupational categories (managers, employees, workers) at the scale of 79 IRIS. Segregation is measured using three indicators covering distinct dimensions: the dissimilarity index (evenness), the isolation and interaction indices (exposure) and the location quotient (spatialisation), supplemented by local spatial autocorrelation (LISA).

The results show that the most separated categories are managers and workers, with a relatively low level of segregation. Integrating the social stock does not leave segregation unchanged, but it does not modify all its dimensions in the same way: it slightly accentuates the dissimilarity between managers and workers and the isolation of workers, while marginally diffusing their presence towards new sectors (fall in spatial autocorrelation, new over-representation zones). This observation raises the question of its specificity to Brest. The same method, applied to Marseille and Paris, shows that evenness and exposure move in the same direction, but that the social stock there spatially consolidates groups in their usual sectors, unlike in Brest. These modifications therefore depend not only on the volume of social housing, but also on its location.

Keywords: *segregation, residential segregation, social housing, Brest Métropole, dissimilarity index, LISA*

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INTRODUCTION

For several decades, housing has held a central place in French urban policy. In a context of growing metropolisation, medium-sized cities such as Brest face specific challenges regarding social mix and the distribution of populations across their territory. Yet these intermediate-sized cities remain less documented in the academic literature, whereas large metropolises such as Lyon, Paris or Marseille have already been the subject of numerous studies on residential segregation.

In France, social housing plays an ambivalent role in the making of the city. On the one hand, it is a means of guaranteeing access to housing for the most modest households and of promoting social mix, in line with the objectives of the SRU law (*Solidarité et Renouvellement Urbains*)[1], adopted in December 2000, which significantly altered housing and planning law in order to seek social balance and to remedy the shortage of social housing. On the other hand, its concentration in certain sectors or neighbourhoods can paradoxically feed residential segregation, by grouping the most precarious households in geographically circumscribed areas.

At the same time, the process of gentrification, first observed in large urban centres, now seems to affect all French cities. This phenomenon of revalorisation of certain working-class neighbourhoods attracts better-off populations, which can displace the most vulnerable residents and accentuate spatial inequalities.

This work belongs to this context. It analyses the relationships between the socio-urban organisation of Brest and the distribution of the social housing stock, in order to identify the role, either mitigating or reinforcing, that the latter plays in the residential segregation of the Brest metropolitan area.

In general terms, residential segregation refers to the tendency of population groups to concentrate in distinct urban spaces within the same territory. It can result from constrained mechanisms, but also from the choices of affluent households seeking to live among themselves. In many French cities, the question of segregation is closely associated with that of social housing, whose spatial distribution depends directly on how it is managed and where it is located.

It is from this perspective that we examine the effect of the distribution of the social housing stock on residential segregation: does it contribute to mitigating or to reinforcing it, in a context of urban transformation marked in particular by gentrification? To answer, we compare, at the

[1] "... The SRU law, reinforced by law 2013-61 of 18 January 2013, also aims to improve social mix in large cities and to remedy the shortage of social housing. According to the law, 'municipalities with more than 3,500 inhabitants – and 1,500 inhabitants in Île-de-France – belonging to conurbations or inter-municipal bodies of more than 50,000 inhabitants including at least one municipality of more than 15,000 inhabitants must have 25% of social housing, relative to main residences, by 2025' (Ministère de la Cohésion des territoires, 2017). Some mayors have done more than drag their feet in applying this provision (Sérafini, 2018)". Géoconfluences (translated from French).

scale of 79 IRIS, the segregation of three socio-occupational categories (managers, employees, workers) without and then with the residents of the social housing stock, using three complementary indicators: the dissimilarity index, the isolation and interaction indices, and the location quotient, supplemented by an analysis of local spatial autocorrelation (LISA).

This approach leads us first to draw on a body of literature situated at the intersection of urban sociology and geography, then to present the data and the methodology adopted, before discussing the results obtained in order to provide a clear answer to the research question.

I. REVISITING THE NOTION OF RESIDENTIAL SEGREGATION

1. Definitions

Residential segregation is defined by Oberti and Préteceille (2016) as the unequal distribution of social groups among the sectors of a city. For Fol and Vallée (2023), it can refer to the social homogeneity of certain spaces and to the separation of certain social groups in space, within the same territory. The notion covers delicate and multidimensional realities: it can result from personal choices, in particular on the part of the most affluent households who seek to group together in a logic of self-segregation, as well as from processes of relegation that assign the most modest households to the least valued neighbourhoods of the city.

The theoretical foundations of the study of segregation and spatial differentiation go back to the 1920s, with the thinkers of the Chicago School, based on the study of American cities and more particularly of Chicago. For Louis Wirth (1928)[2], segregation is above all the result of migrations that led to the grouping of populations of the same origins in community sectors, which act as zones of transition and gradual integration. This reading was later challenged, notably by Massey and Denton (1993)[3]. In *American Apartheid*, they show that segregation, far from being a transitory state, tends to reproduce and strengthen itself over time, generating cumulative negative effects for the populations that experience it. In France, the question has long been structured around working-class suburbs and large social housing estates, which prompted a debate on the existence of French-style ghettos. Loïc Wacquant (2006) responds to it in *Parias urbains*: starting from a comparison between American black ghettos and French housing estates, he demonstrates that "urban marginality is not woven from the same fabric everywhere" and that French banlieues cannot be equated with ghettos in the American sense of the term.

Regarding the causes of segregation, Oberti and Préteceille (2016) identify institutional logics, stemming from public allocation and construction policies, individual logics, linked to households' residential strategies, and economic logics, linked to the functioning of the housing market. For Fol and Vallée (2023), these logics can be complementary and are not mutually exclusive: segregation always results from an accumulation of factors whose relative weight varies according to the city, the period and the context. Wacquant (2006) for his part stresses the role of the State, whose planning and housing policies participate very actively in creating and maintaining the spatial marginality of the most disadvantaged households.

[2]Reported by Fol and Vallée in: Fol, Sylvie, and Julie Vallée. 2023. *Ségrégation*. Mobidic.

[3]Also reported by Fol and Vallée in the same work.

2. Measuring residential segregation

From the mid-twentieth century onwards, the quantitative measurement of residential segregation developed considerably. In 1955, in "A Methodological Analysis of Segregation Indexes", published in the *American Sociological Review*, Duncan and Duncan proposed the first major measurement tool: the dissimilarity index, whose definition is given later.

Seeking to measure segregation properly in American cities, Massey and Denton proposed to consider it as a multidimensional phenomenon rather than as a simple distance between groups. In 1988, they organised the various indices into five broad dimensions: evenness, exposure, concentration, clustering and centralisation[4]. This framework remains the reference in the literature, even if, for reasons of parsimony set out in the methodological section, this work uses only a selection of indicators.

The work of Philippe Apparicio is a considerable contribution to the calculation of segregation indices. It builds on Massey and Denton's five dimensions and adds a spatial and local dimension. He lists several indices for each of these dimensions and develops an automated calculation tool integrated into MapInfo[5].

Apparicio shows that, depending on the index used, the same segregation phenomenon can give rise to very different measurements. It is therefore important to choose indicators according to the dimensions one wishes to analyse, and to favour their complementarity rather than their accumulation. Concretely, a group may be well distributed across a city (low dissimilarity index) while residing in spatially contiguous neighbourhoods (strong spatial autocorrelation). Only the complementarity of indicators measuring distinct dimensions therefore yields a complete analysis of a territory's segregation.

3. Social housing stock and segregation

The link between social housing and residential segregation is the subject of a debate at the heart of urban policy. Between the 1950s and the 1970s, the construction of large housing estates on the urban periphery played a major role in concentrating the social housing stock in a few well-defined sectors of French cities. In their analysis, Botton et al.[6] show that, on average, the segregation index of the social housing stock stood at 61% in large French urban units in 1990, making it the most segregated housing category. The SRU law aims to correct these inequalities: between 1990 and 2015, this index fell from 61% to around 50% thanks to the construction of social housing in areas that had none. For Botton et al., however, this decline

[4]Massey, Douglas S., and Nancy A. Denton. 1988. "The Dimensions of Residential Segregation." *Social Forces* 67 (2): 281.

[5]Apparicio, Philippe. 2000. "Les indices de ségrégation résidentielle : un outil intégré dans un système d'information géographique." *Cybergeo*.

[6]Botton, Hugo, Pierre-Yves Cusset, Clément Dherbécourt, and Alban George. 2020. "Quelle évolution de la ségrégation résidentielle en France ?" *La note d'analyse* no. 92 (4): 1-12.

was more marked before the SRU law than after it, which qualifies its effectiveness. In their view, the law essentially targets segregation between municipalities, by requiring social housing in those that lacked it, and not segregation between neighbourhoods of the same territory. Despite this decline, the social housing stock remains more segregated than the private rental sector (34%) or owner-occupation (33%). Finally, the study shows that, over the same period, the share of the most modest households living in social housing increased: workers and employees, who made up 29% of social housing residents in 1990, made up 33% in 2015. These households thus live increasingly in a stock that does evolve in space but remains far more segregated than the others. The social housing stock and residential segregation are therefore closely linked.

In sum, this literature leads us to understand residential segregation in the Brest metropolitan area as the result of cumulative dynamics: urban policies that tend to concentrate the social housing stock in certain sectors, allocation logics that continue to steer the most precarious families towards them, and neighbourhood effects that produce social inequalities.

II. METHODOLOGICAL APPROACH

1. Data

The analysis draws on the "Individus localisés au canton-ou-ville" file from the 2022 population census carried out by INSEE. For each surveyed individual, it provides a set of socio-demographic variables together with information on their household and main residence, including their socio-occupational category, their location and whether or not they belong to the social housing stock. Location is given at the canton-or-city scale, with a precision that can go down to the IRIS for areas of at least 200 inhabitants. Since this information comes from statistical surveys, calculations must be weighted by the individual weight (IPONDI), which yields counts close to reality during aggregation. It must therefore be used with its decimals retained in order to avoid rounding errors.

Another point of attention concerns counts within an area: those above 500 inhabitants can normally be used with confidence, whereas those below 200 must be used with caution, as they may not be meaningful owing to the imprecision of the survey. In our case, this file makes it possible to reconstruct the spatial distribution of socio-occupational categories and whether or not they belong to a social housing organisation.

Study area

As indicated in the introduction, large cities such as Paris, Marseille or Lyon, although they remain highly relevant for such an analysis, have already been the subject of an abundant literature on residential segregation. The choice of study area therefore fell on a medium-sized city, under-represented in existing work on our theme. The leading conurbation of western Brittany and the third in the region behind Nantes and Rennes, with just over 213,400 inhabitants, the Brest metropolitan area is not necessarily chosen for the particular relevance of the site to the research question, but through a deliberate, considered and assumed choice: that of a little-known territory whose socio-residential composition could not be anticipated. This absence of prior knowledge is precisely what makes the approach valuable, as it is exploratory and free of pre-existing representation biases.

The metropolitan area is made up of eight municipalities, with Brest as the central city, itself divided into seven administrative districts. The scale of analysis corresponds to the 79 housing-type IRIS distributed across all the municipalities of Brest Métropole.

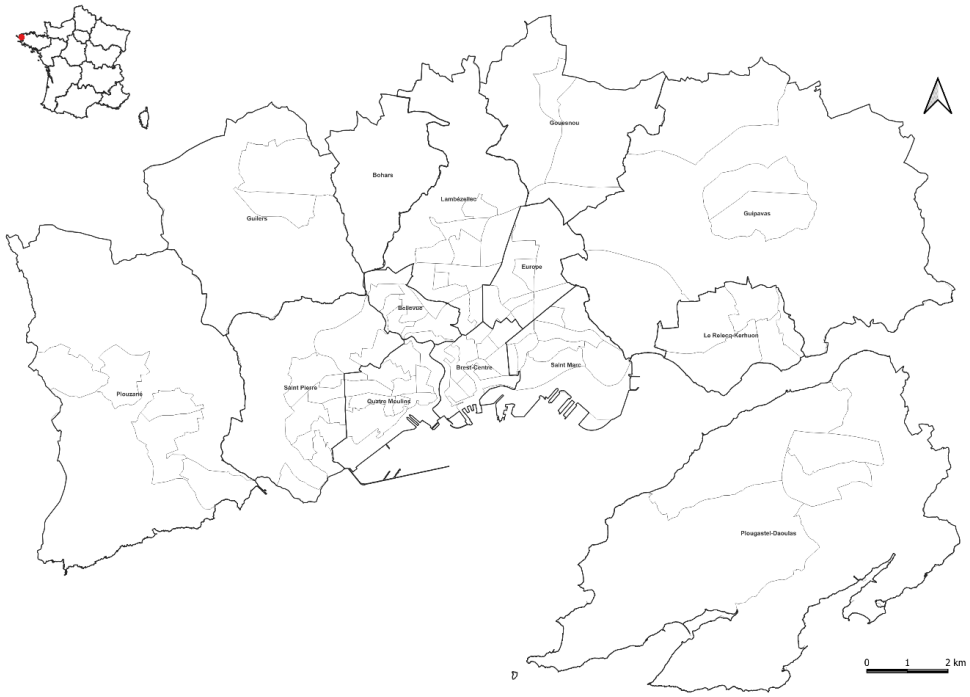


Figure 1. Location of the study area, 2026

Population studied

This choice rests first on the results of several studies, notably Botton et al. (2020), which shows that managers are one and a half times more unevenly distributed than workers and employees, and reveals a growing tendency towards self-segregation among managers. The INSEE Analyses Provence-Alpes-Côte d'Azur study (2024) also indicates that the socio-occupational categories with the most extreme incomes display the strongest segregation, the intermediate categories being better distributed in space. Managers and workers, located at the two ends of the hierarchy, are therefore a priori the categories for which the effects of residential segregation are most marked. As for employees, Botton et al. (2020) observe a clear rise in their share in the social housing stock; they thus complete this trio, which allows the link between social housing and residential segregation to be observed precisely.

These three categories also offer better statistical coverage at the IRIS scale, which is necessary for a robust calculation of the indices, and allow a direct comparison between an upper pole (managers) and a working-class pole (employees and workers) of the social structure. Finally, they are quite simply the socio-occupational categories most used in empirical studies.

2. Method

There are several ways of measuring the level of segregation of a territory, notably distance-based methods or classification methods (hierarchical clustering, k-means, etc.). Our analysis rests on the calculation of residential segregation indices, within the multidimensional framework proposed by Massey and Denton (1988) and developed by Philippe Apparicio

(2000). These indices fall into single-group indices, which measure the distribution of a group relative to the whole reference population, and inter-group indices, which analyse the distribution of one group relative to another.

Indices belonging to the same dimension often yield redundant results, without providing additional information. To avoid this pitfall and to refocus the analysis on the mechanisms most directly interpretable with respect to our research question, this work retains only three indicators, each covering a distinct dimension of segregation, supplemented by a local spatial analysis:

- **evenness**, measured by the dissimilarity index (D), which refers to the unequal distribution of one group, or of two groups relative to each other, among the spatial units of the territory;
- **exposure**, measured by the isolation index (xPx) and its inter-group counterpart, the interaction index (xPy), which refer to the probability of contact between members of the same group or of different groups within the same spatial unit;
- **spatialisation**, measured by the location quotient (LQ), which locates the over- or under-representation of a group in each spatial unit, and supplemented by the analysis of local spatial autocorrelation (LISA maps), which identifies significant clusters of neighbouring units.

In a second stage, and in order to answer the research question, the analysis rests on the comparison of two scenarios that make it possible to isolate the effect of the social housing stock on residential segregation:

(a) the first scenario excludes from the analysis individuals belonging to a social housing organisation;
(b) the second reintegrates them, in order to obtain a result covering the housing stock as a whole.

Comparing the results of the two scenarios makes it possible to observe the effect of the social housing stock on the spatial distribution of the selected socio-occupational categories. The exercise relies on two automated Python pipelines: the first filters, reorganises and aggregates the "Individus localisés au canton-ou-ville" file at the IRIS scale; the second carries out the complete segregation analysis, from the calculation of the indices to mapping.

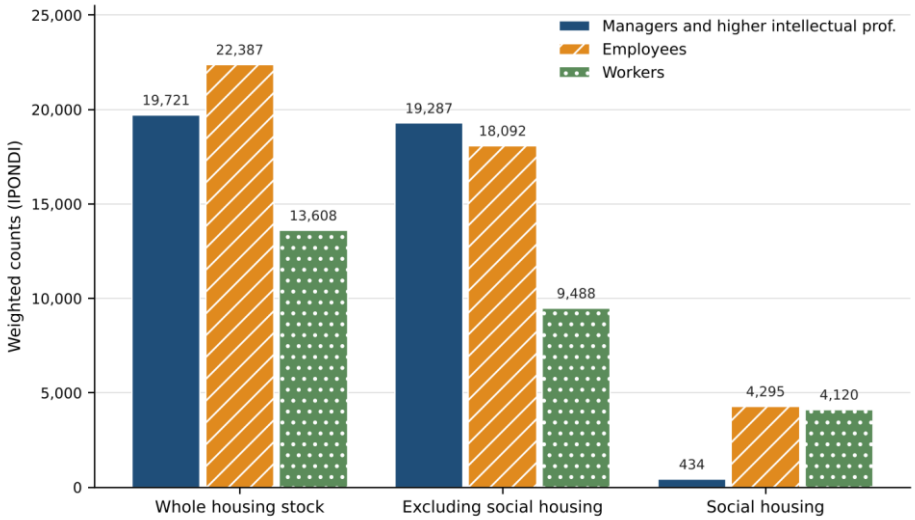
2.1. Filtering, reorganising and aggregating the data

This first step consists in filtering and aggregating the raw data of the detailed file in order to build the table used to calculate the indices. The work carried out in Python comprises the following steps: (1) loading the detailed file from Google Drive; (2) filtering, by IRIS, the individuals of the study area; (3) selecting the useful columns (CANTVILLE, IRIS, CATIRIS, GS, HLML and IPONDI); (4) filtering the GS variable (socio-occupational group) to keep only

categories 3, 5 and 6, corresponding respectively to managers and higher intellectual professions, employees and workers; (5) replacing the GS codes with their labels.

(6) The selected individuals are then grouped by IRIS and by GS by summing IPONDI, which gives the actual count of each category per IRIS, presented in columns: one row per IRIS and one column per category (managers, employees, workers). At this stage, only housing-type (H) IRIS are retained: the other types (A and D) are very sparsely populated and provide no relevant information on residential segregation, and excluding them avoids biasing the results.

To ease reading, (7) the label of each IRIS is added, then (8) the area of each IRIS is calculated by joining with the "CONTOUR-IRIS.SHP" spatial layer from the data.gouv.fr platform. Finally, (9) the result is exported in Excel and GeoJSON formats. For the scenario excluding the social housing stock, an additional step is added upstream: only individuals for whom HLML = 2 (not belonging to a HLM organisation) are retained. In what follows, the reference population of an IRIS is the sum of the counts of the three selected categories.



Graph 1. Numbers of managers, employees and workers after filtering and aggregation, "housing" IRIS only

The housing stock as a whole is mostly composed of employees and managers, while workers remain a minority. This configuration is clearly reversed in the social housing stock: managers are almost absent (about 2.2% of their numbers), whereas they are the majority in the private stock and in owner-occupation, while workers make up 30.3% of their numbers, followed by employees (about 19%).

2.2. Measuring residential segregation

2.2.1. Segregation indices used

Once filtered and aggregated, the data serve as the basis for the segregation analysis. In line with the choice set out above, two dimensions are measured by indices: evenness and exposure. All the indices are calculated using the "segregation" library of the Python Spatial Analysis Library (PySAL).

a. Evenness

Dissimilarity index (D) (inter-group index): it compares the distributions of two groups X and Y among the spatial units and measures their spatial separation. It ranges from 0 (identical distributions) to 1 (maximum dissimilarity) and expresses the share of either group that would have to change spatial unit for the two to obtain identical distributions (Apparicio, 2000).

$$D(X, Y) = \frac{1}{2}\sum_i \left| \frac{x_i}{X} - \frac{y_i}{Y} \right|$$

where x*i* and y*i* are the counts of groups X and Y in spatial unit i, and X and Y their respective total counts in the study area.

When Y represents the whole reference population minus group X, the index is calculated in its single-group form D(X), which measures the distribution of a group relative to the rest of the population (Apparicio, 2000):

$$D(X) = \frac{1}{2}\sum_{i=1}^{n} \left| \frac{x_i}{X} - \frac{t_i - x_i}{T - X} \right|$$

where x*i* is the count of group X in spatial unit i, X its total count in the study area, t*i* the reference population of spatial unit i and T the total reference population.

b. Exposure

Isolation index (xPx) (single-group index): it measures the probability that a member of group X shares their spatial unit with another member of the same group. It ranges from 0 to 1, the maximum value meaning that the group is totally isolated (Apparicio, 2000).

$$xP_x = \sum_{i=1}^{n} \left(\frac{x_i}{X} \times \frac{x_i}{t_i} \right)$$

where x*i* is the count of group X in spatial unit i, X its total count in the city and t*i* the reference population of spatial unit i.

Interaction index (xPy) (inter-group index): it refers to the probability that a member of group X shares their spatial unit with a member of group Y (Apparicio, 2000). This index is not symmetric: the probability that a manager meets a worker is not the same as the probability that a worker meets a manager. Exposure indices are sensitive to the demographic weight of the groups, which must be taken into account when interpreting the results.

$$xP_y = \sum_{i=1}^{n} \left(\frac{x_i}{X} \times \frac{y_i}{t_i} \right)$$

where x*i* and y*i* are the counts of groups X and Y in spatial unit i, X the total count of group X in the city and t*i* the reference population of spatial unit i.

These indices give global values for the whole territory. They must therefore be supplemented by a mapping of segregation at the scale of each IRIS, which the calculation of the location quotient and the analysis of local spatial autocorrelation (LISA) make possible (Apparicio, 2000; Apparicio and Gelb, 2025; Lee and Kim, 2025).

2.2.2. Spatialisation of segregation

a. Location quotient

The location quotient (LQ) is calculated for each spatial unit. It makes it possible to locate the over- or under-representation of a group in the city, by relating the proportion of the group in the spatial unit to its proportion in the city. If LQ = 1, the share of group X in the spatial unit equals that of the city; if LQ > 1, the group is over-represented in the unit, and conversely if LQ < 1. This measure can be mapped.

$$LQ_i = \frac{x_i / t_i}{X / T}$$

where x*i* is the count of group X in spatial unit i, t*i* the total population of spatial unit i, X the count of the group in the city and T the total population of the city.

Once the LQ has been calculated and mapped, the task is to observe whether units with a high (or low) quotient are clustered in the city, and then to locate these clusters. This requires first calculating the global Moran's I, then calculating and mapping the LISA. The LQ thus serves as the common thread of the entire spatial analysis.

b. Spatial autocorrelation

Unlike the location quotient, which treats spatial units independently, Moran's and LISA indices consider each IRIS in relation to its neighbours, which requires defining the neighbourhood. Among the possible methods, Queen contiguity was chosen: it considers as neighbours two spatial units sharing a common border or at least one point of their border (vertex or edge), unlike Rook contiguity, which requires a common border. The weights matrix w*ij* equals 1/n*i* if i and j are neighbours, and 0 otherwise (n*i* being the number of neighbours of i). In Python, it is computed via libpysal.weights.Queen.from_dataframe().

Because of the choice to retain only housing-type IRIS (79 out of 93), the matrix reveals six disjoint connected components corresponding to the peripheral sectors of the metropolitan area, notably Plougastel-Daoulas, separated from the other municipalities by the sea, as well as sectors in the west, which include many IRIS of another type. This does not affect the calculation, since the esda library natively handles disconnected components, but it calls for vigilance in interpretation.

Moran's I (global Moran)

Global Moran measures whether spatial units with high or low LQ are statistically clustered, dispersed or randomly distributed in the metropolitan area. It quantifies the degree of similarity or differentiation of neighbouring spatial units.

$$I = \frac{n}{S_0} \cdot \frac{\sum_i \sum_j w_{ij} (z_i - \bar{z}) (z_j - \bar{z})}{\sum_i (z_i - \bar{z})^2}$$

where n is the number of spatial units, w*ij* the spatial weight between units i and j (Queen contiguity), z*i* the location quotient of unit i, z̄ the mean LQ over the whole territory and S_0 the sum of all spatial weights.

- if I > 0: positive autocorrelation, similar units are close together;
- if I < 0: negative autocorrelation, neighbouring units tend to be different;
- if I = 0: no autocorrelation.

As in many studies, significance is assessed by a p-value calculated from 999 permutations[7], with a seed of 12345. The result is significant if the p-value is less than or equal to 0.005. The calculation is carried out with the esda library (esda.Moran()) of the PySAL ecosystem.

Local Moran's I (LISA)

Introduced in 1995 by Luc Anselin, local indicators of spatial association (LISA) complement global Moran's I by identifying the spatial units where autocorrelation is strong, weak or not significant.

$$I_i = \frac{z_i - \bar{z}}{m_2} \sum_j w_{ij} (z_j - \bar{z}), \qquad m_2 = \frac{\sum_i (z_i - \bar{z})^2}{n}$$

where z*i* is the location quotient of unit i, z̄ the mean LQ, w*ij* the spatial weight between units i and j, n the number of IRIS and m_2 the variance of the z*i* over the whole study area.

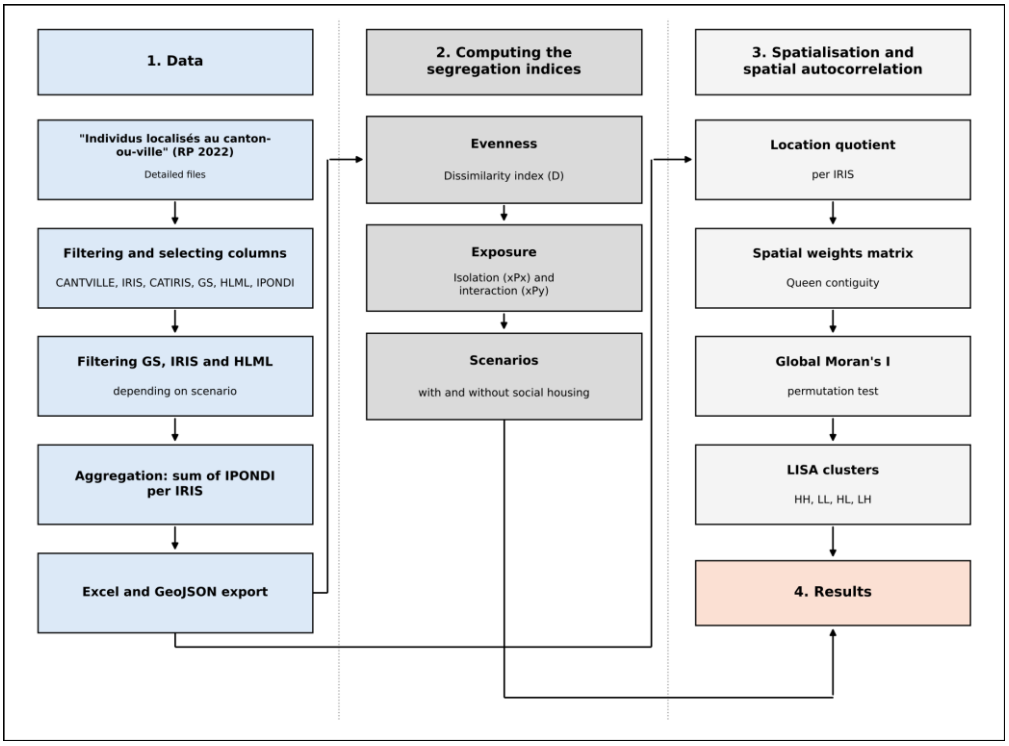
The result provides one value per spatial unit, then classified according to its LQ and that of its neighbours:

- High-High (HH): high-value IRIS surrounded by high-value neighbours;
- Low-Low (LL): low-value IRIS surrounded by low-value neighbours;
- High-Low (HL): high-value IRIS surrounded by low-value neighbours (enclave);

[7]The LQ values are randomly reassigned to the IRIS 999 times, the global I index is recalculated at each draw, and the p-value corresponds to the proportion of permutations giving a Moran's I as extreme as the one observed, in order to check whether the observed spatial autocorrelation is significant or due to chance.

- Low-High (LH): low-value IRIS surrounded by high-value neighbours (enclave);
- not significant (ns): p-value > 0.05.

The calculation is carried out with esda.Moran_Local(), also with 999 permutations, a 5% threshold and a seed of 12345, the classification being obtained via cluster_labels(). All the maps and diagrams are produced with the matplotlib library.



Graph 2. Methodological diagram

III. ANALYSIS OF THE RESULTS

Before analysing the results of the segregation indicators, it is useful to place the Brest metropolitan area in its socio-residential context. There are different types of social housing depending on the financing mode. PLAI housing ("Prêt Locatif Aidé d'Intégration") is intended for the most disadvantaged households, with the lowest income ceilings and rents in the social stock. PLUS housing ("Prêt Locatif à Usage Social") is the most widespread form of financing and corresponds to the majority of eligible households. PLS housing ("Prêt Locatif Social") is aimed at middle-income households exceeding the PLUS ceilings, with slightly higher rents. Finally, PLI housing ("Prêt Locatif Intermédiaire"), which offers the highest income ceilings and rents, forms the link between the social stock and the private sector. Our analysis does not take this typology into account and treats the social stock as a whole; it is nevertheless relevant to map it in order to get an overall view of the structure of the stock in the metropolitan area. This socio-residential description relies mainly on data from ADEUPa (*Portrait social de Brest Métropole*, 2024), supplemented by a map based on INSEE data.

Only four in ten households are owners in Brest, compared with at least seven in ten in the other municipalities of the metropolitan area, which is distinguished by a strong presence of private and social housing. About 16,988 households lived in the social stock in Brest (18.3% of households) in 2020; in the other municipalities, this rate ranges from 7.6% in Plougastel-Daoulas to 14.9% in Le Relecq-Kerhuon. The map of the share of social housing by IRIS (Figure 2) confirms this distribution at the intra-municipal scale: the social stock is clearly concentrated in the IRIS of the Bellevue district, but also in some IRIS of the Europe, Saint-Pierre, Quatre Moulins districts and the south of Lambézellec. Conversely, the city centre and the peripheral municipalities have the least. The IRIS with a high share of social housing in Brest correspond essentially to the priority neighbourhoods of urban policy (QPV), which have not been mapped here.

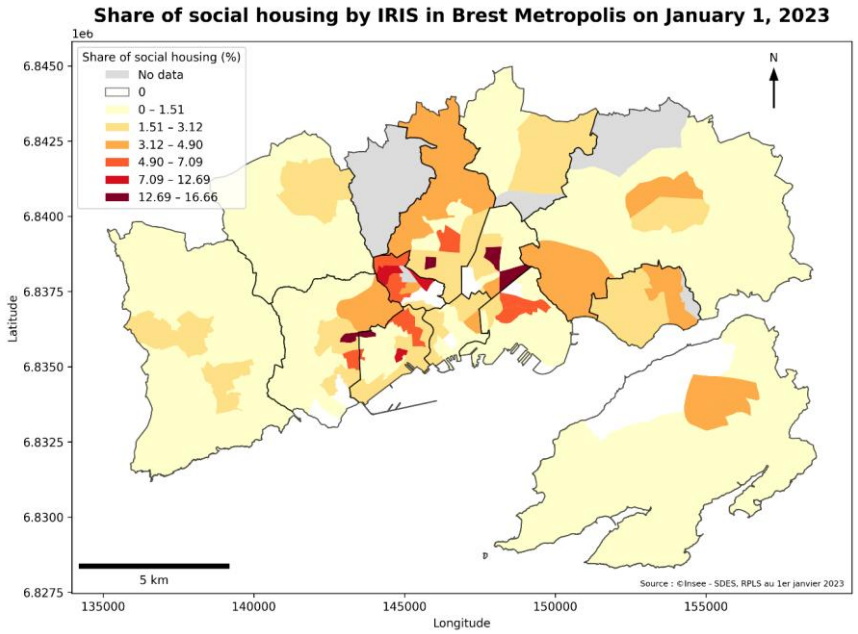


Figure 2. Share of social housing stock by IRIS in Brest Métropole, 2023

The map of the dominant financing by 100-metre grid cell (Figure 3) further clarifies this configuration.

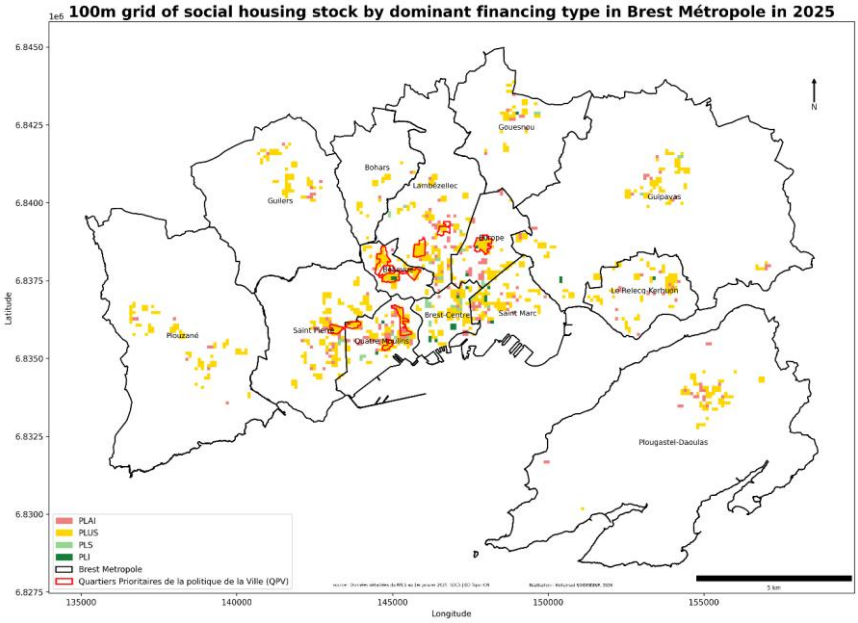


Figure 3. 100 m grid of the social housing stock by dominant financing type, Brest Métropole, 2025

PLUS largely dominates throughout the metropolitan area, while PLAI is scattered across the whole city, including in the peripheral municipalities. PLS and PLI are mainly scattered in the south of the metropolitan area, in particular in Brest-Centre, Quatre Moulins and Saint Marc.

1. Residential segregation in Brest excluding the social housing stock

This first scenario concerns the residential segregation of the three socio-occupational categories, excluding tenants of the social housing stock.

Table 1. Dissimilarity (D, group versus rest of the population) and isolation (xPx) indices, excluding social housing, Brest Métropole, 2022

Index	Managers	Employees	Workers
D (group / rest)	0.252	0.169	0.158
xPx	0.462	0.410	0.224

The results reveal a hierarchy between the categories. On the evenness dimension, managers show the highest value (D = 0.252), which means they are the most unevenly distributed group among the IRIS of the metropolitan area, ahead of employees (0.169) and workers (0.158). This result confirms the study by Botton et al. (2020), already drawn on earlier in this work, according to which managers are distributed more unevenly than workers and employees, owing to a tendency towards self-segregation among upper-class households. The value nevertheless remains moderate: about a quarter of managers would have to change IRIS for their distribution to be identical to that of the rest of the population.

The isolation index confirms this geography. With xPx = 0.462 for managers, a manager has about a 46.2% probability of sharing their IRIS with another manager, compared with only a 22.4% chance for a worker of sharing theirs with another worker. This result is no doubt partly due to group size, but it clearly reflects a tendency towards self-segregation among managers. Employees occupy an intermediate position (0.410), close to that of managers, which reflects their higher demographic weight.

Table 2. Dissimilarity index D(X, Y) between groups, excluding social housing

D	Managers	Employees	Workers
Managers	—	0.249	0.264
Employees	0.249	—	0.110
Workers	0.264	0.110	—

For comparisons between groups, dissimilarity is highest between managers and workers (D = 0.264, a relatively low value compared with other study areas), followed by the managers–employees pair (0.249). It is markedly lower between employees and workers (0.110). These values reflect an opposition between managers and the working classes that is more marked than within the latter.

Table 3. Interaction index xPy (row: exposed group; column: group encountered), excluding social housing

xPy	Managers	Employees	Workers
Managers	—	0.442	0.299
Employees	0.471	—	0.337
Workers	0.608	0.643	—

The interaction index shows that a manager has about a 30% probability of meeting a worker in their IRIS and 44% of meeting an employee: they are therefore more distant from workers than from employees. Conversely, a worker has a 64.3% chance of meeting an employee in their IRIS and 60.8% of meeting a manager. This result is a direct consequence of the numbers involved: since the interaction index is sensitive to demographic weight, workers, a minority in our sample, live in spaces where the other categories are more numerous, which mechanically makes their probability of meeting someone outside their group high.

Spatialisation of segregation excluding social housing

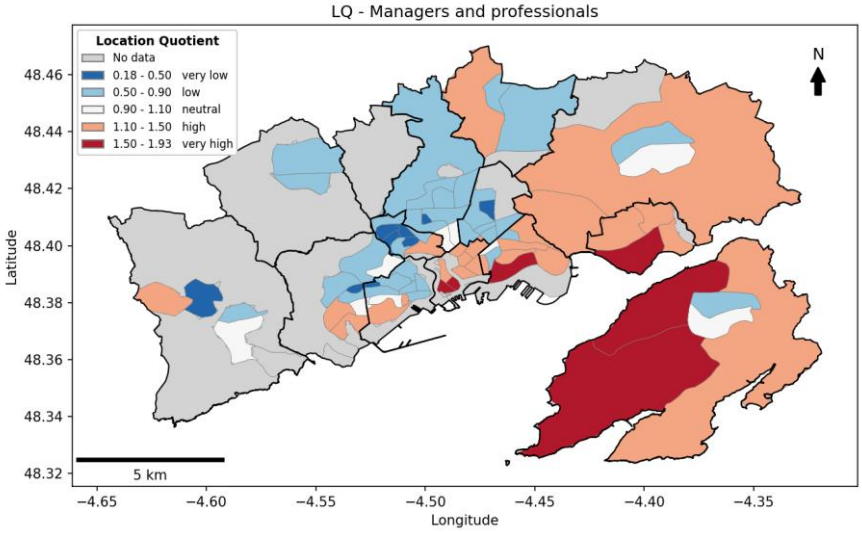


Figure 4. Location quotient of managers, excluding social housing, Brest Métropole, 2022

The location quotient of managers outside the social stock (Figure 4) reveals a clear fragmentation between the east/south-east and the north/west of the metropolitan area. The IRIS with the highest LQ are located in Plougastel-Daoulas and Le Relecq-Kerhuon, as well as in the Brest-Centre and Saint Marc districts. Conversely, the lowest values are found in the IRIS of the Bellevue and Lambézellec districts, but also in some QPV, notably Kerourien (Saint Pierre/Quatre Moulins) and Pontanezen (Europe), which form compact zones of under-representation of managers.

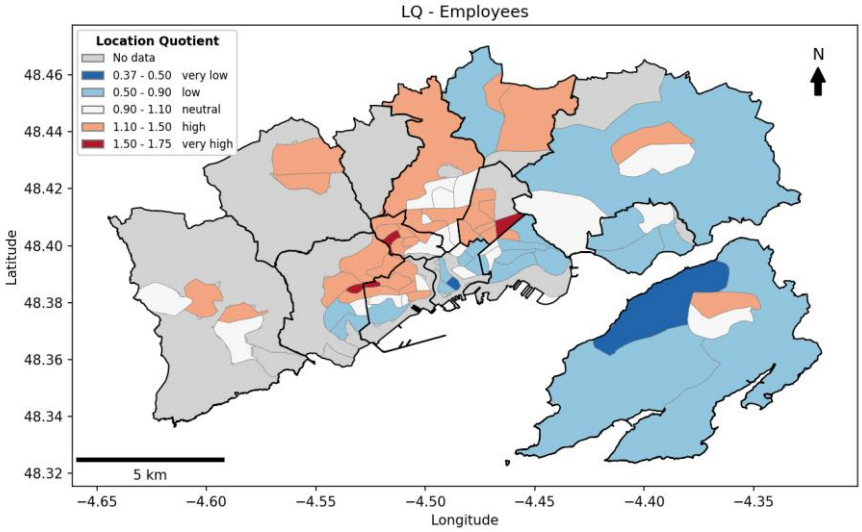


Figure 5. Location quotient of employees, excluding social housing, Brest Métropole, 2022

For employees, the configuration is almost the inverse of that of managers. The strongest over-representation is located in the north and west of Brest, in the Bellevue and Saint Pierre/Quatre Moulins districts and the south of the Europe district, with values up to 1.75. A moderate over-representation (1.10 to 1.50) crosses the limits of the city of Brest and reaches peripheral municipalities such as Guilers, Gouesnou, Guipavas or Plougastel-Daoulas, which nevertheless include IRIS where employees are under-represented (minimum LQ of 0.37), creating enclaves. Employees are also strongly under-represented in the city centre.

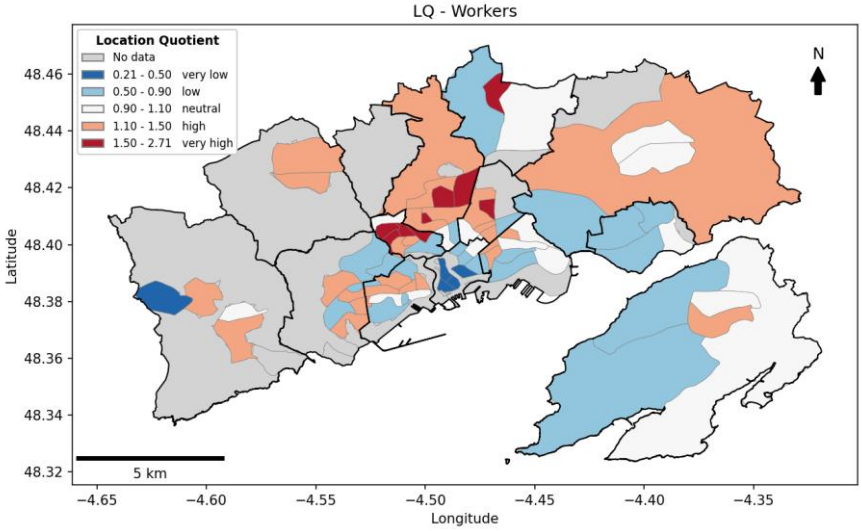


Figure 6. Location quotient of workers, excluding social housing, Brest Métropole, 2022

The LQ map of workers (Figure 6) shows a strong over-representation in the IRIS of Bellevue and Lambézellec, in the centre of the Europe district (Pontanezen QPV) and in the north of the municipality of Gouesnou, with maximum values reaching 2.71, the highest of all categories. Conversely, Brest-Centre shows a clear under-representation of workers (minimum LQ), which confirms the absence of impoverishment of the city centre. Outside the social stock, segregation in Brest thus draws a working-class pole in the north and west (Bellevue,

Lambézellec, etc.) and privileged coastal areas, where workers are clearly under-represented, in the south and east (Brest-Centre, Saint Marc, Plougastel-Daoulas, Le Relecq-Kerhuon and south-west Guipavas).

Table 4. Global Moran's I, excluding social housing

	Managers	Employees	Workers
Moran's I	0.523	0.462	0.301
p-value	0.001	0.001	0.002
Significant (0.005)	Yes	Yes	Yes

Moran's I gives a positive and significant spatial autocorrelation for the three groups. IRIS with a high (or low) LQ are therefore not distributed at random: they tend to cluster. The same hierarchy is found in the intensity of this autocorrelation: managers come first (0.523), then employees (0.462) and finally workers (0.301). The residential geography of managers is thus spatially more marked than that of employees and workers.

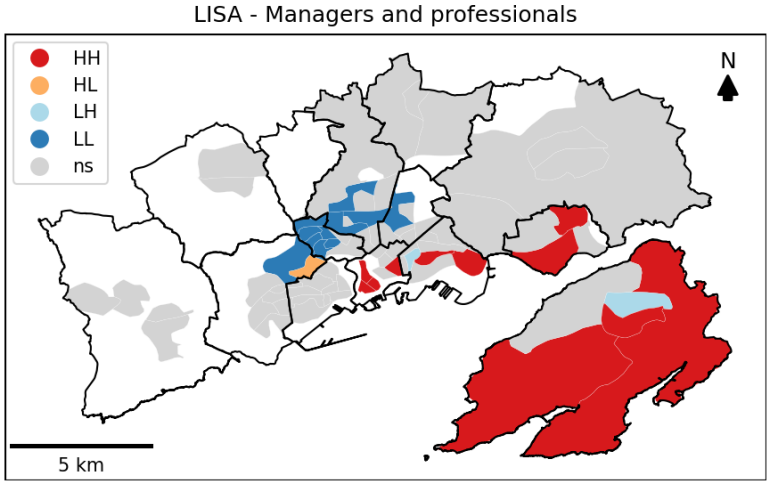


Figure 7. LISA of managers, excluding social housing, Brest Métropole, 2022

Figure 7 shows a vast High-High cluster covering almost the whole municipality of Plougastel-Daoulas, in the south, followed by more modest clusters in Le Relecq-Kerhuon and in the southern districts of Brest (Saint Marc and Brest-Centre). The High-High clusters are thus concentrated along the eastern façade of the metropolitan area. At the opposite end, in the north-west of Brest, from the north-east of Saint Pierre to the heart of Bellevue and the south of Lambézellec, passing through the west of Europe, are the Low-Low clusters, with a High-Low enclave on the northern border between Saint Pierre and Quatre Moulins (a high-LQ IRIS of managers surrounded by low-LQ neighbours). Two Low-High enclaves also appear in the centre of Plougastel-Daoulas and in Saint Marc, corresponding to low-LQ IRIS surrounded by high-LQ neighbours.

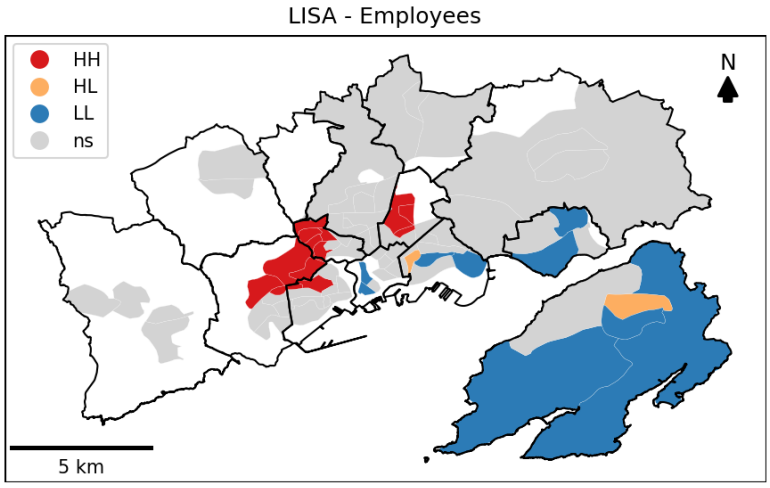


Figure 8. LISA of employees, excluding social housing, Brest Métropole, 2022

The LISA of employees broadly reproduces the inverse geography of that of managers. Figure 8 shows a large High-High cluster in the west of Brest (Saint Pierre, Quatre Moulins and Bellevue) and a second, made up of three IRIS, in the Europe sector. Almost all the High-High clusters of managers have become Low-Low clusters of employees, and the Low-High enclaves of managers have become High-Low enclaves of employees, that is, IRIS with a high LQ of employees surrounded by low-LQ neighbours.

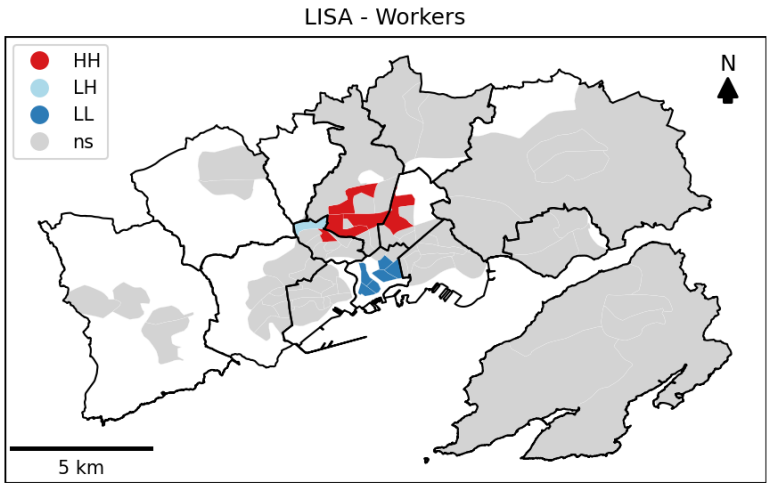


Figure 9. LISA of workers, excluding social housing, Brest Métropole, 2022

The LISA of workers offers a reading different from the other two. A single, compact High-High cluster is observed, concentrated around Lambézellec, to which a Low-High IRIS is added. The Low-Low clusters are located in Brest-Centre, the city centre being historically a space mainly composed of managers.

This first scenario, which excludes social housing, reveals a relatively low level of segregation for all categories: moderate dissimilarity, a higher probability of intragroup contact among managers, and a spatial autocorrelation that is more marked for them than for employees and workers. This reading draws a geography opposing a working-class pole concentrated north of

the city centre, a more or less homogeneous intermediate pole in the west (Quatre Moulins, Saint Pierre) and a privileged pole occupying the eastern façade, from the centre to the east of the metropolitan area (Brest-Centre, Plougastel-Daoulas, Saint Marc, etc.). The next step is to reintegrate the social housing stock in order to observe any changes.

2. Residential segregation for the housing stock as a whole

Table 5. Dissimilarity (D, group versus rest of the population) and isolation (xPx) indices, including social housing, Brest Métropole, 2022

Index	Managers	Employees	Workers
D (group / rest)	0.280	0.162	0.173
xPx	0.423	0.423	0.280

Reintegrating the social housing stock reveals contrasting changes depending on the category. On the evenness dimension, the D index rises slightly for managers, from 0.252 to 0.280 (+0.028), and for workers, from 0.158 to 0.173 (+0.015), whereas it falls very slightly for employees, from 0.169 to 0.162 (−0.007).

The isolation index rises for the working-class categories, from 0.224 to 0.280 (+0.056) for workers and from 0.410 to 0.423 (+0.013) for employees, whereas it falls for managers (from 0.462 to 0.423). By including the social stock, a worker therefore has a higher chance of sharing their IRIS with another worker (28%), which reflects a slight intensification of the concentration of the working classes in their own spaces. The fall in managers' isolation is mechanically explained by their near-absence from the social stock (2.2% of their numbers, see Graph 1): adding social tenants increases the reference population of the IRIS concerned without changing the number of managers, which dilutes their probability of rubbing shoulders with other managers.

Table 6. Dissimilarity index D(X, Y) between groups, including social housing

D	Managers	Employees	Workers
Managers	—	0.272	0.299
Employees	0.272	—	0.128
Workers	0.299	0.128	—

The results show a slight rise in dissimilarity for all three pairs: from 0.264 to 0.299 for the managers–workers pair, from 0.249 to 0.272 for managers–employees and from 0.110 to 0.128 for employees–workers. The most notable rise is that of the managers–workers pair (from 26% to 30%), which reflects a strengthening of the socio-residential separation between affluent and working classes. These variations nevertheless remain very modest and in no way imply a large-scale fragmentation.

Table 7. Interaction index xPy (row: exposed group; column: group encountered), including social housing

xPy	Managers	Employees	Workers
Managers	—	0.475	0.348
Employees	0.419	—	0.368
Workers	0.504	0.605	—

The interaction indices of managers with the other groups increase (from 0.442 to 0.475 with employees and from 0.299 to 0.348 with workers). These results confirm one of the effects of the social stock in the Brest metropolitan area: it distributes employees and workers across almost all IRIS and, by spreading them into spaces where managers already live, logically increases the probability that the latter meet them there. Conversely, the probability that a worker or an employee meets a manager in their IRIS falls, by 0.104 for workers (0.608 to 0.504) and by 0.052 for employees (0.471 to 0.419). The working-class categories therefore have a lower probability of interacting with the upper categories in their own spaces, which is consistent with the strengthening of their isolation.

Spatialisation of segregation including social housing

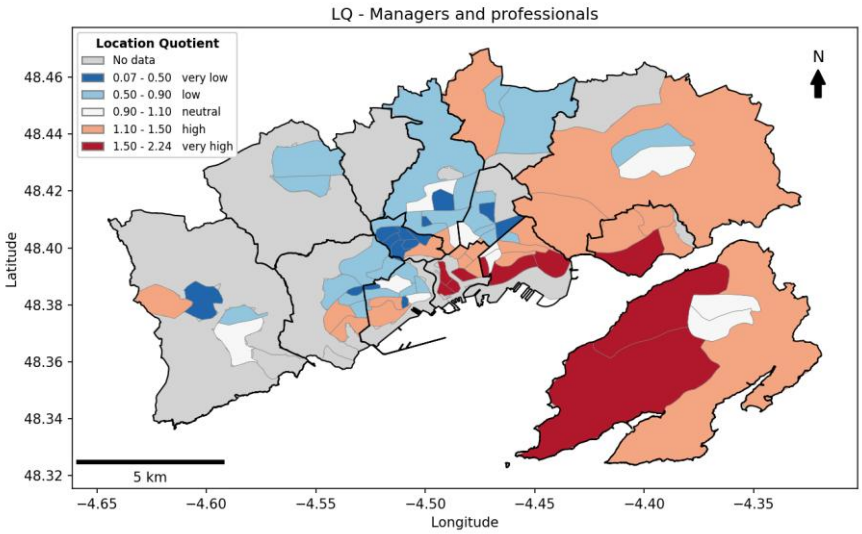


Figure 10. Location quotient of managers, including social housing, Brest Métropole, 2022

The geography is almost identical to that of managers outside the social stock, with a rising maximum quotient (from 1.93 to 2.24), which reflects a slightly more marked polarisation. The municipalities of Guipavas, Plougastel-Daoulas and Le Relecq-Kerhuon, as well as the Saint Marc and Brest-Centre districts, remain the over-representation sectors, whereas Lambézellec and Bellevue remain those where managers are least present. A few changes are worth noting: new over-representation zones appear in the IRIS of the heart of Brest-Centre and its surroundings, and new under-representation IRIS appear in the Bellevue sector, which goes in the direction of the previous indices and reflects a strengthening of affluent households in their own spaces.

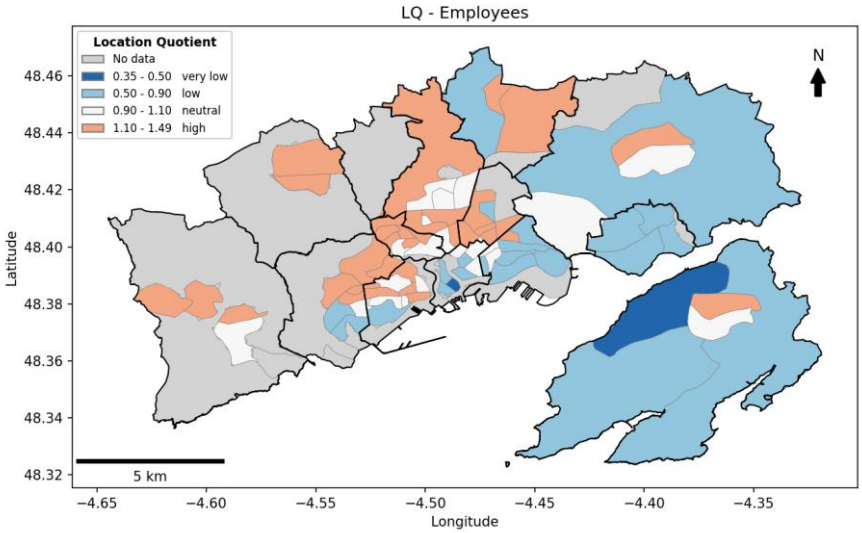


Figure 11. Location quotient of employees, including social housing, Brest Métropole, 2022

The geography of employees has changed little. Over-representation remains in the districts already mentioned in the first scenario (Saint-Pierre, Bellevue and Lambézellec), but no IRIS is any longer very strongly over-represented (fall in the maximum LQ compared with the first scenario). The integration of the social stock is therefore accompanied by a slight reduction in the maximum representation of employees, consistent with the near-stability of their isolation (0.410 to 0.423) and the slight fall in their dissimilarity.

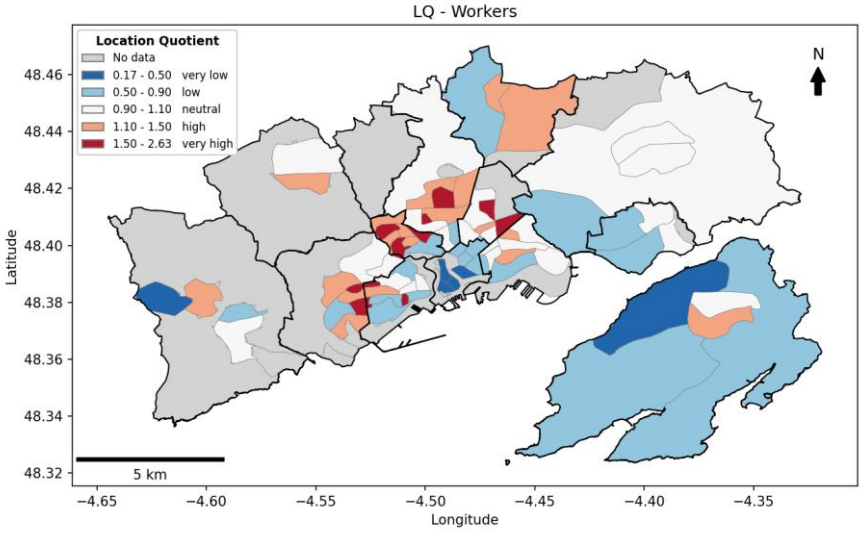


Figure 12. Location quotient of workers, including social housing, Brest Métropole, 2022

Figure 12 shows a redistribution of workers towards neutral zones or zones sparsely populated by this category. Besides the usual sectors (Bellevue and Lambézellec), which are reinforced, a new sector of very strong over-representation appears in Saint Pierre, with under-represented IRIS moving to neutral representation, or even over-representation: this is the Kerourien QPV, mainly composed of social housing. The LQ of the north of Lambézellec and of the east of the metropolitan area decreases. Including workers who are tenants of the social stock increases

the proportion of workers in the metropolitan area as a whole, but this rise is uneven: it is concentrated in certain sectors of the city of Brest, in particular Bellevue, the south of Lambézellec and Saint Pierre, whereas the north of Lambézellec and the east see their LQ decrease.

Table 8. Global Moran's I, including social housing

	Managers	Employees	Workers
Moran's I	0.501	0.485	0.253
p-value	0.001	0.001	0.004
Significant (0.005)	Yes	Yes	Yes

The variations in global Moran's I depend on the category. It rises slightly for employees, from 0.462 to 0.485, whereas it falls for managers (from 0.523 to 0.501) and for workers (from 0.301 to 0.253). These three indices still reflect a positive and statistically significant spatial autocorrelation. The fall in workers' autocorrelation, even as their dissimilarity and isolation rise, may seem contradictory, but it remains consistent: the social stock is spread over several IRIS, notably on the periphery, in addition to the usual sectors of Lambézellec and Bellevue, which reduces the overall clustering measured by Moran's I, while at the same time accentuating their concentration within these historic sectors, hence the rise in the dissimilarity and isolation indices.

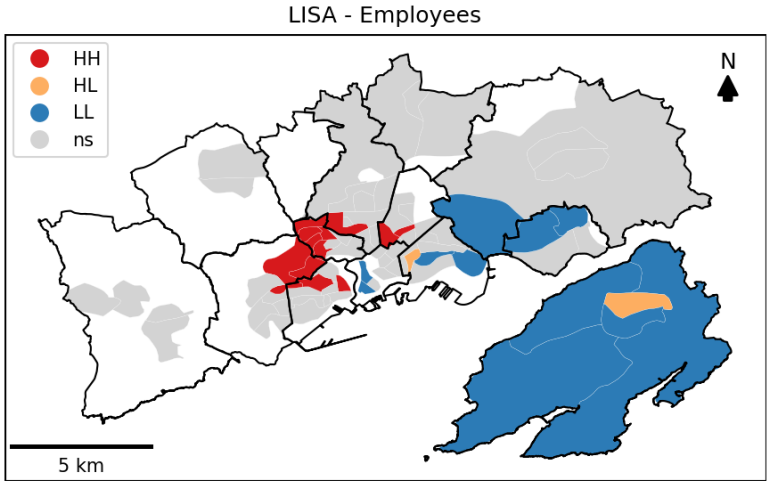


Figure 13. LISA of managers, including social housing, Brest Métropole, 2022

The spatial structure remains almost identical to that of the first scenario. The High-High clusters are located in the heart of Brest-Centre and in the sectors along the south-eastern façade of the metropolitan area, with a few adjustments: the Low-High point of Saint Marc disappears to join the High-High cluster already in place, and the High-Low IRIS of the north-west of Saint Pierre, which was becoming High-High, moves to the south of Bellevue, becoming an IRIS with a strong presence of managers surrounded by IRIS with a low presence of managers.

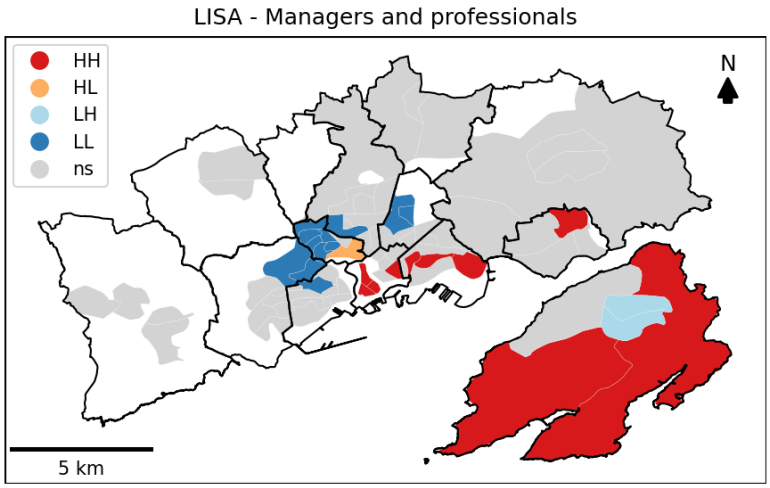


Figure 14. LISA of employees, including social housing, Brest Métropole, 2022

For employees, the composition also remains broadly stable. The High-High cluster of north-west Brest is maintained, with a few modifications: it loses one IRIS in the Saint Pierre sector but gains two others, one in Quatre Moulins and the other in Lambézellec. The High-High cluster of the Europe district also evolves. The Low-Low cluster of Plougastel-Daoulas extends to almost the whole municipality, surrounding a High-Low IRIS; that of Le Relecq-Kerhuon also extends to the southern IRIS of Guipavas, while that of Brest-Centre remains stable.

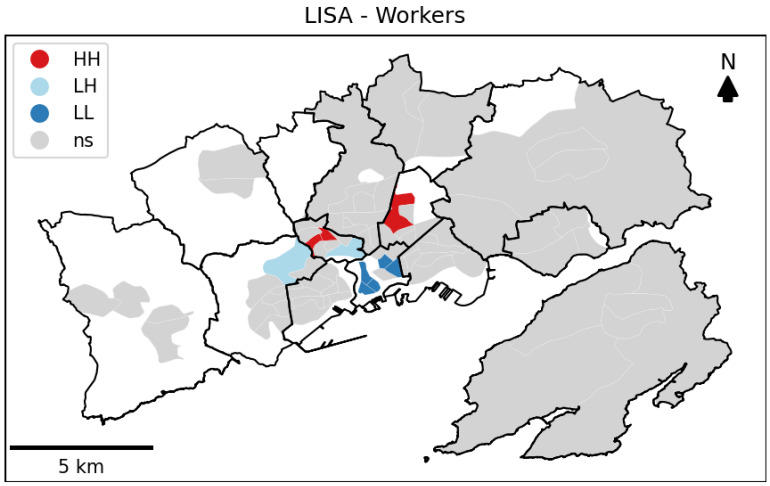


Figure 15. LISA of workers, including social housing, Brest Métropole, 2022

Conversely, the High-High cluster of workers in the Bellevue-Lambézellec sector shrinks appreciably compared with the first scenario: all the IRIS of Lambézellec leave it, leaving only a small core in the heart of Bellevue, to which another High-High IRIS is added. The Europe sector remains stable. Two Low-High points appear, one in the south of Bellevue and the other in the north-west of Saint Pierre, on the border with Bellevue. The latter forms an enclave of IRIS with a low presence of workers surrounded by neighbours with a strong working-class presence, which is logical since to the north lies the Bellevue sector, historically composed mainly of

workers, and to the south the Kerourien QPV, between Saint Pierre and Quatre Moulins, where a large share of workers also lives. The Low-Low clustering of the city centre is maintained.

3. Effect of the social housing stock on residential segregation in the Brest metropolitan area

The set of results allows a first observation: the level of residential segregation of the three socio-occupational categories in the Brest metropolitan area is relatively low at the IRIS scale, judging by the values of the indices. It is nevertheless sufficient to answer our research question, which is to determine whether the distribution of the social stock mitigates or amplifies residential segregation. Table 9 sets the two scenarios side by side for each of the three dimensions analysed.

Table 9. Change in segregation indicators between the two scenarios (with minus without social housing), Brest Métropole, 2022

Dimension	Indicator	Without social housing	With social housing	Change
Evenness	D managers – workers	0.264	0.299	+0.035
	D managers – employees	0.249	0.272	+0.023
	D employees – workers	0.110	0.128	+0.018
	D managers / rest	0.252	0.280	+0.028
	D employees / rest	0.169	0.162	−0.007
	D workers / rest	0.158	0.173	+0.015
Exposure	xPx managers	0.462	0.423	−0.039
	xPx employees	0.410	0.423	+0.013
	xPx workers	0.224	0.280	+0.056
	xPy managers → employees	0.442	0.475	+0.033
	xPy managers → workers	0.299	0.348	+0.049
	xPy employees → managers	0.471	0.419	−0.052
	xPy employees → workers	0.337	0.368	+0.031
	xPy workers → managers	0.608	0.504	−0.104
	xPy workers → employees	0.643	0.605	−0.038
Spatialisation	Moran's I managers	0.523	0.501	−0.022
	Moran's I employees	0.462	0.485	+0.023
	Moran's I workers	0.301	0.253	−0.048

Note: changes calculated from the rounded values of the previous tables. For xPy, the first group is the exposed group, the second the group encountered.

These results show that the social housing stock does not leave segregation unchanged, but that it does not alter it uniformly: certain dimensions are transformed, in different directions and with different intensities.

Evenness. Dissimilarity rises for all three pairs of categories, more clearly between managers and workers (+0.035), and the single-group D index rises for managers (+0.028) and workers (+0.015), while falling slightly for employees (−0.007). Taking the social stock into account thus slightly accentuates the separation between managers and the working classes, in particular workers. This result runs counter to what one might expect of the social stock: by giving modest households access to sectors they could not occupy in the private stock or in owner-occupation, one might hope that it would play a role of socio-residential mix. The social stock remains the most segregated of the different housing types, and it reinforces the concentration of the majority categories it houses, in this case workers[8].

Exposure. The changes are asymmetric. The isolation of workers rises markedly (+0.056) and that of employees slightly (+0.013), whereas that of managers falls (−0.039), owing to their near-absence from the social stock. Interaction probabilities move in the same direction: managers are more exposed to employees (+0.033) and to workers (+0.049), whereas the latter are less exposed to managers (−0.052 for employees, −0.104 for workers). The social stock thus redistributes the potential for contact to the detriment of the working-class categories, who meet fewer managers in their own IRIS, even as it spreads their presence into IRIS where managers live. These variations, sensitive to the demographic weight of the groups, must be read with caution.

Spatialisation. The effect here is less clear-cut. The spatial autocorrelation of workers falls (−0.048), as does that of managers (−0.022), whereas that of employees rises (+0.023). The LQ and LISA maps clarify this movement: the working-class presence spreads beyond its usual sectors (new over-representation zone in Saint Pierre, in the Kerourien QPV, fall in the LQ in the north of Lambézellec and in the east, shrinking of the High-High cluster of Lambézellec), while strengthening in certain sectors (Bellevue, south of Lambézellec, Saint Pierre). The Low-High enclaves of workers that appeared in the south of Bellevue and the north-east of Saint Pierre also show that the social stock does not only produce homogeneous spaces, but also generates enclaves at the sub-district scale, where IRIS of more affluent private housing persist in sectors otherwise dominated by social housing. On the managers' side, the rise in the maximum LQ (from 1.93 to 2.24) and the appearance of new under-representation IRIS in Bellevue reflect a slightly increased polarisation.

Finally, the set of results reveals a characteristic specific to certain French cities such as Brest: the absence of impoverishment of the city centre. In both scenarios, the centre of Brest remains

[8]Botton, Hugo, Pierre-Yves Cusset, Clément Dherbécourt, and Alban George. 2020. "Quelle évolution de la ségrégation résidentielle en France ?" *La note d'analyse* no. 92 (4): 1-12.

a Low-Low cluster for the working-class categories and an over-representation zone for the upper categories, in particular managers. Residential segregation in Brest Métropole rests on a fairly homogeneous working-class pole in the north-west of the city centre, made up of Bellevue, Lambézellec, Guilers, etc., reinforced by the Saint Pierre sector in the second scenario, and on a vast privileged space comprising the city centre and the east-south-east façade of the metropolitan area (Le Relecq-Kerhuon, Plougastel-Daoulas, etc.). The social stock does not correct this geography: it reconfigures it at the margin, consolidating it on certain dimensions and diffusing it on others.

4. Application to other territories and contexts: Marseille and Paris

As announced in the methodology, this section tests whether the effect observed in the Brest metropolitan area generalises or differs according to the urban context, by applying the same method. To keep the exposition manageable, only the single-group indicators (D and xPx indices) and the spatialised indicators (LQ, Moran's I, LISA) are analysed for these two study areas.

Marseille

Table 10. Dissimilarity (D, group versus rest), isolation (xPx) indices and Moran's I, Marseille, 2022

Indicator	Managers	Employees	Workers
D (group / rest), excluding social housing	0.337	0.185	0.280
D (group / rest), including social housing	0.385	0.187	0.285
xPx, excluding social housing	0.491	0.429	0.262
xPx, including social housing	0.462	0.458	0.289
Moran's I, excluding social housing	0.713	0.453	0.614
Moran's I, including social housing	0.736	0.523	0.726

In Marseille, segregation is more marked for managers (D = 0.337, compared with 0.280 for workers and 0.185 for employees), as is isolation (xPx = 0.491), that is, a 49% probability that a manager meets another manager in their space, ahead of employees (0.429) and workers (0.262).

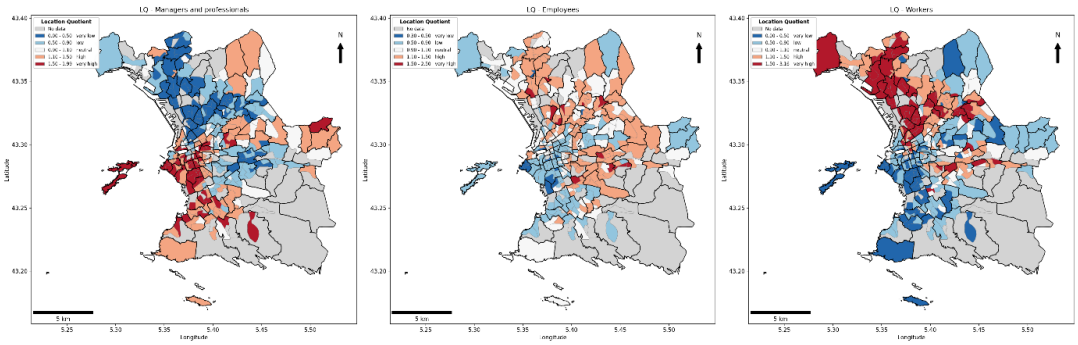


Figure 16. Location quotient, excluding social housing, Marseille, 2022

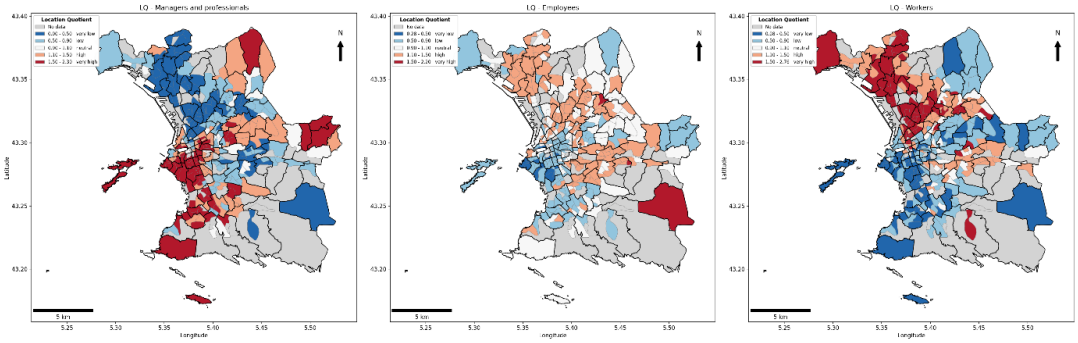


Figure 17. Location quotient, including social housing, Marseille, 2022

The maps show Marseille's classic divide between a north over-represented mainly by the working classes and a south largely composed of managers.

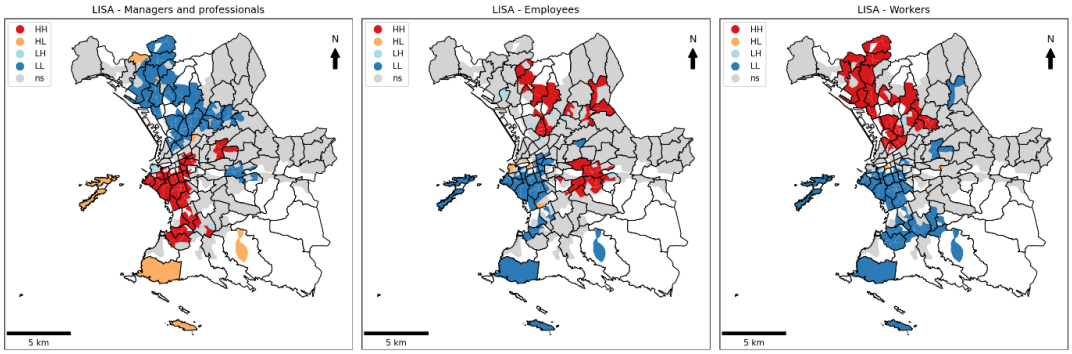


Figure 18. LISA, excluding social housing, Marseille, 2022

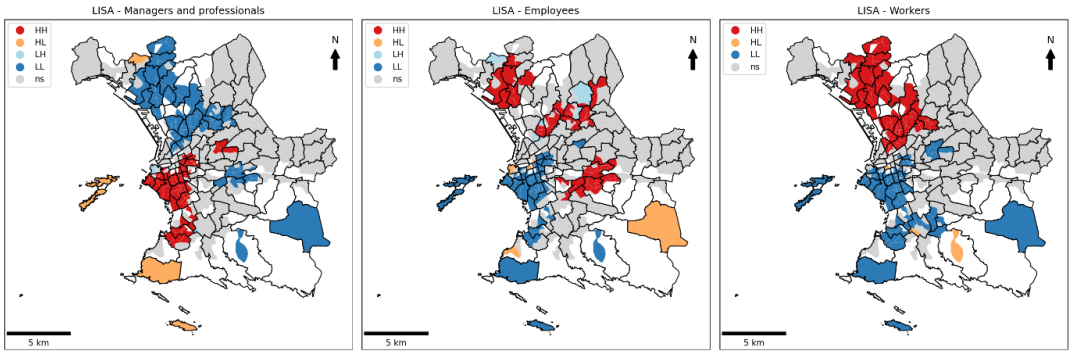


Figure 19. LISA, including social housing, Marseille, 2022

The old city centre concentrates both a High-High cluster of managers and a High-High cluster of workers. The integration of the HLM stock tends to accentuate this geography slightly: the dissimilarity of managers rises (from 0.337 to 0.385), as does the isolation of employees (from 0.429 to 0.458) and workers (from 0.262 to 0.289), whereas that of managers falls (from 0.491 to 0.462). This change is not accompanied by a dispersal in space: spatial autocorrelation rises for the three groups (from 0.713 to 0.736 for managers, from 0.453 to 0.523 for employees and from 0.614 to 0.726 for workers). The rise in the isolation of workers and employees, combined with that of Moran's I, reflects a local consolidation of these groups in their traditional sectors, as the LISA maps show.

Paris

Table 11. Dissimilarity (D, group versus rest), isolation (xPx) indices and Moran's I, Paris, 2022

Indicator	Managers	Employees	Workers
D (group / rest), excluding social housing	0.170	0.153	0.244
D (group / rest), including social housing	0.275	0.230	0.291
xPx, excluding social housing	0.794	0.185	0.079
xPx, including social housing	0.737	0.272	0.117
Moran's I, excluding social housing	0.429	0.352	0.299
Moran's I, including social housing	0.533	0.481	0.492

In Paris, it is workers who show the highest dissimilarity (D = 0.244), with a much less marked isolation than the other groups, managers displaying the highest xPx value (0.794), that is, nearly an 80% probability of sharing one's IRIS with another manager. This gap, largely linked to the demographic weight of the groups in the capital (the isolation index is very sensitive to it), clearly distinguishes Paris from the other two study areas.

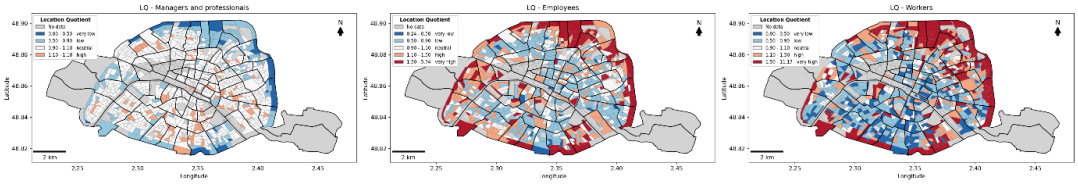


Figure 20. Location quotient, excluding social housing, Paris, 2022

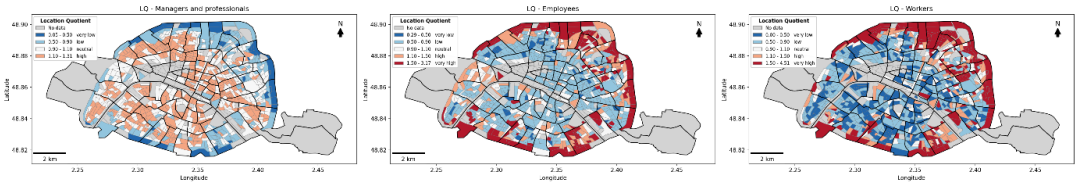


Figure 21. Location quotient, including social housing, Paris, 2022

The spatialisation shows an opposition between the far north and the far east, dominated by workers, a minority in the city, and by employees, and the heart of the city, mainly composed of managers.

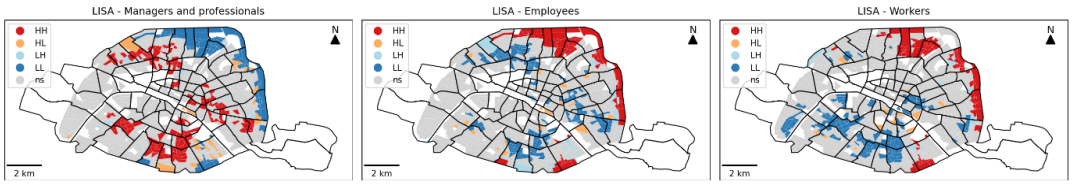


Figure 22. LISA, excluding social housing, Paris, 2022

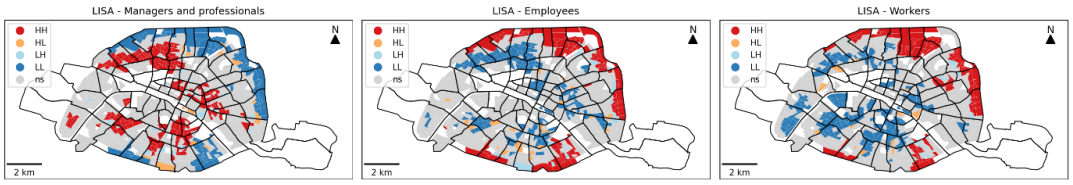


Figure 23. LISA, including social housing, Paris, 2022

The arrival of HLM residents accentuates this divide: dissimilarity rises for the three groups (+0.105 for managers, +0.077 for employees, +0.047 for workers), the isolation of employees (from 0.185 to 0.272) and workers (from 0.079 to 0.117) increases, as does spatial autocorrelation (from 0.429 to 0.533 for managers, from 0.352 to 0.481 for employees and from 0.299 to 0.492 for workers, the latter experiencing the largest change, +0.193). Since the social stock is mainly located in the far east and the north of the city, this strengthening is accompanied by an increased presence of workers and employees in these sectors.

Summary

Table 12. Change in indicators between the two scenarios (with minus without social housing) in Brest, Marseille and Paris

Indicator	Brest	Marseille	Paris
D (group / rest), managers	+0.028	+0.048	+0.105
D (group / rest), employees	−0.007	+0.002	+0.077
D (group / rest), workers	+0.015	+0.005	+0.047
xPx, managers	−0.039	−0.029	−0.057
xPx, employees	+0.013	+0.029	+0.087
xPx, workers	+0.056	+0.027	+0.038
Moran's I, managers	−0.022	+0.023	+0.104
Moran's I, employees	+0.023	+0.070	+0.129
Moran's I, workers	−0.048	+0.112	+0.193

The results show different levels of segregation from one context to another, but the underlying pattern remains the same: managers and workers are the categories most unevenly distributed in space. Table 12 allows the dimensions to be compared. On evenness and exposure, the three cities move in the same direction: the dissimilarity of managers and workers rises, the isolation of workers and employees rises, and that of managers falls. Reintegrating the social stock therefore does not reduce the level of segregation on either of these two dimensions, but slightly accentuates it. It is on the spatial dimension that trajectories diverge: spatial autocorrelation falls in Brest for managers and workers, whereas it rises for all groups in Marseille and Paris.

In other words, in Brest, the social stock accompanies a marginal diffusion of the working-class presence towards new sectors, whereas in Marseille and Paris it consolidates groups in their usual spaces, the rise in autocorrelation being particularly marked in Paris. These results suggest that the effect of the social stock on residential segregation probably depends not only

on the number of social housing units within cities, but also on their location: depending on whether it is located in sectors that are already disadvantaged or in more affluent neighbourhoods, it takes part in different spatial dynamics.

IV. DISCUSSION

The answer to our research question is nuanced: the distribution of the social housing stock does not mitigate residential segregation in the Brest metropolitan area, but it modifies certain of its dimensions, in different directions. It slightly accentuates the uneven distribution of the categories (dissimilarity) and the isolation of the working classes (exposure), while marginally diffusing their presence in space (spatialisation). The social stock thus behaves neither as a simple aggravating factor nor as a corrective tool: it reconfigures segregation more than it reduces it.

On the evenness dimension, dissimilarity rises for all pairs of categories, and more clearly between managers and workers. This finding contradicts the objectives of public social housing and social mix policies, in particular the SRU law, which aimed at greater socio-residential heterogeneity of sectors by allowing modest households to access spaces they could not occupy in the private stock. It does, however, join the general finding of several studies, notably that of Botton et al. (2020), according to which the social stock remains more segregated than the other stocks.

On the exposure dimension, the isolation of workers increases while that of managers declines: workers meet more workers in their IRIS and fewer managers, whereas managers, present in IRIS to which the social stock now brings employees and workers, are more exposed to them. This asymmetry shows that the spatial diffusion of the social stock does not translate into a symmetric increase in contact between groups: the apparent mix at the scale of the metropolitan area is paid for, by the working-class categories, with lower exposure to the upper categories in their own spaces. Since these indices are sensitive to the demographic weight of the groups, this reading must nevertheless be handled with caution.

On the spatialisation dimension, the effect is the most ambivalent. On the one hand, the social stock tightens the concentration of modest households, in this case workers, in their historic sectors (Bellevue, Lambézellec) and brings out new over-representation sectors (Saint Pierre). On the other, the fall in the spatial autocorrelation of workers, the shrinking of their High-High cluster and the fall in their LQ in certain IRIS, in particular in the east (Guipavas and Gouesnou), reveal a marginal diffusion of their presence towards other sectors. Outside the social stock, workers moreover appear more concentrated in certain spaces (the east, for example, on the LQ map); their reintegration modifies this geography. Finally, in both scenarios, the city centre of Brest remains an affluent sector made up of private housing, mainly occupied by managers (35% of the local population, compared with 15% for workers and employees[9]).

[9]Docquois, Barbara. 2024. "Portrait social de Brest métropole (2024)." ADEUPa Brest Bretagne, Observatoire, November 2024. Brest Métropole.

Putting the method into perspective with other cities confirms that the social stock modifies segregation according to configurations specific to each territory. In Brest, Marseille and Paris, dissimilarity and isolation move in the same direction; but spatial autocorrelation, which falls in Brest for managers and workers, rises everywhere in Marseille and Paris, where the social stock consolidates groups in their usual spaces. The intensity and form of these modifications thus vary according to the socio-urban context, and seem to depend as much on the location of the social stock as on its volume. The results therefore call for reflection, by planning and social mix policies, not only on the volume of social housing, which tends to consolidate existing residential concentrations, but also on its spatial distribution.

Finally, these results call for caution: the differences between the two scenarios in Brest Métropole remain small, in a city where the level of residential segregation is relatively low, and the choice of three indicators does not claim to exhaust all the dimensions of segregation. The answer nevertheless remains clear: the distribution of the social housing stock is far from being a tool for correcting the segregation and residential inequalities produced by the market; it rather contributes to reconfiguring, and on some points to maintaining, a geography inherited from former public housing policies.

CONCLUSION

The aim of this work was to measure, using three indicators of residential segregation (dissimilarity index, isolation and interaction indices, location quotient) supplemented by a LISA analysis, and applied to two scenarios, without and then with the residents of the social housing stock, the effect of social housing on residential segregation in the Brest metropolitan area.

Two main findings emerge. The first is that the social geography of Brest is organised according to an inverted centre–periphery structure: in the north, a working-class pole composed mainly of workers and employees, extending towards the west of Brest and of the metropolitan area, is opposed to a privileged pole in the south, from the city centre to the west and along the south-eastern façade, mainly composed of managers and higher intellectual professions. The second is that the social housing stock modifies certain dimensions of residential segregation without changing its overall structure: it slightly accentuates the uneven distribution of the categories, in particular between managers and workers, reinforces the isolation of workers and reduces their exposure to managers, while marginally diffusing their presence towards other sectors, as indicated by the fall in spatial autocorrelation and the appearance of new over-representation zones. The social stock therefore does not mitigate segregation, but reconfigures it.

Applying the method to Marseille and Paris confirms that these modifications go broadly in the same direction for evenness and exposure, but that their spatial dimension, and hence their intensity, depends on the urban context: in Marseille and Paris, the social stock is accompanied by a spatial consolidation of groups, unlike in Brest.

This work could nevertheless have been deepened by integrating the temporal dimension. It would have been more relevant to draw on two different dates and compare the results in order to observe how the mechanism evolves over time. If, in our analysis, the social housing stock does not seem to have a mitigating effect on residential segregation in the Brest metropolitan area, the question remains whether this result was the same twenty years ago: it is possible that its intensity is no longer the same as at the end of the 1990s and the beginning of the 2000s, and that the trend will reverse in the future. Integrating the temporal dimension thus opens new avenues for research.

Moreover, the scale of analysis adopted (the IRIS) has limits. It would have been more interesting to have a wider intermediate scale, such as is found in cities like Marseille, Paris or Lyon, which have many arrondissements; the seven administrative districts of Brest remain insufficient to constitute a relevant scale. In addition, retaining only housing-type IRIS (79 out of 93) leads to a map with empty spaces and to a neighbourhood matrix (Queen contiguity) made up of six connected components disconnected from one another, even though this did not affect the calculation.

These limits do not, however, call into question the set of results obtained: the social housing stock, far from being a tool for mitigating residential segregation in the Brest metropolitan area, modifies certain of its dimensions and takes part in the consolidation of its geography.

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APPENDICES

Appendix 1. Automated Python code for filtering, aggregating and reorganising the "Individus localisés au canton-ou-ville" file

```
import pandas as pd
import os
import geopandas as gpd


folder_path = '/content/drive/MyDrive/Brest Metropole'
file_path = os.path.join(folder_path, 'FD_INDCVIZC_2022.csv')


try:
    df = pd.read_csv(file_path, sep=';')
    print("Fichier CSV 'FD_INDCVIZC_2022.csv' chargé avec succès.")
except FileNotFoundError:
    print(f"Erreur: Le fichier '{file_path}' n'a pas été trouvé.")
    exit()
except Exception as e:
    print(f"Une erreur est survenue lors du chargement du fichier: {e}")
    exit()


chemin_couche_iris = os.path.join(folder_path, 'CONTOUR-IRIS-Brest Metropole.shp')
try:
    gdf_iris = gpd.read_file(chemin_couche_iris)
    print("Fichier Shapefile 'CONTOUR-IRIS-Brest Metropole.shp' chargé avec succès.")
except FileNotFoundError:
    print(f"Erreur: Le fichier '{chemin_couche_iris}' n'a pas été trouvé.")
    exit()
except Exception as e:
    print(f"Une erreur est survenue lors du chargement du Shapefile: {e}")
    exit()


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    290190168, 292120101, 292120102, 292120103, 292120104, 292120105, 292120106,
    290610101, 290610102, 290610103, 290610104, 290690101, 290690102, 290690103,
    290750101, 290750102, 290750103, 290750104, 290750105, 291890103, 291890104,
    291890105, 291890106, 291890107, 292350101, 292350102, 292350103, 292350104,
    292350105
]


df_processed_hlml2 = df.copy()
df_processed_hlml2['IRIS'] = df_processed_hlml2['IRIS'].astype(str).str.replace(r'\\.0$', '',
regex=True)
df_processed_hlml2['IRIS'] = pd.to_numeric(df_processed_hlml2['IRIS'],
errors='coerce').astype('Int64')
df_processed_hlml2 = df_processed_hlml2.dropna(subset=['IRIS'])
df_processed_hlml2 = df_processed_hlml2[df_processed_hlml2['IRIS'].isin(iris_codes_to_keep)]
print(f"DataFrame filtré par IRIS. Dimensions: {df_processed_hlml2.shape}")


selected_columns = ['CANTVILLE', 'IRIS', 'CATIRIS', 'GS', 'HLML', 'IPONDI']
df_processed_hlml2 = df_processed_hlml2[selected_columns].copy()
print(f"Colonnes sélectionnées et réorganisées. Dimensions: {df_processed_hlml2.shape}")


df_processed_hlml2 = df_processed_hlml2[df_processed_hlml2['HLML'].astype(str) == '2'].copy()
#filtering for households in social housing (HLML = 2)
print(f"DataFrame filtré par HLML='2'. Dimensions: {df_processed_hlml2.shape}")


gs_values_to_keep = ['3', '5', '6']
df_processed_hlml2 =
df_processed_hlml2[df_processed_hlml2['GS'].astype(str).isin(gs_values_to_keep)].copy()
print(f"DataFrame filtré par 'GS'. Dimensions: {df_processed_hlml2.shape}")


gs_mapping = {
    '3': 'Cadres et professions intellectuelles supérieures',
    '5': 'Employés/Employées',
    '6': 'Ouvriers/Ouvrières'
}
df_processed_hlml2['GS'] = df_processed_hlml2['GS'].replace(gs_mapping)
print("Codes 'GS' remplacés par des libellés.")
```

```
print("Aperçu du DataFrame traité après remplacement des codes 'GS':")
display(df_processed_hlml2.head())
print("Nouvelles fréquences des valeurs 'GS' après remplacement:")
display(df_processed_hlml2['GS'].value_counts(dropna=False))


df_aggregated_hlml2 = df_processed_hlml2.groupby(['IRIS', 'CATIRIS',
'GS'])['IPONDI'].sum().reset_index()


df_pivoted_hlml2 = df_aggregated_hlml2.pivot_table(index=['IRIS', 'CATIRIS'], columns='GS',
values='IPONDI', fill_value=0).reset_index()


df_pivoted_hlml2.columns.name = None


print("DataFrame agrégé par IRIS et CATIRIS avec les effectifs par catégorie socioprofessionnelle
(HLML=2):")


df_habitat_only_hlml2 = df_pivoted_hlml2[df_pivoted_hlml2['CATIRIS'] == 'H'].copy()


print("DataFrame agrégé filtré par IRIS de type 'H' (Habitat) (HLML=2):")
display(df_habitat_only_hlml2.head())
print(f"Dimensions finales du DataFrame filtré par CATIRIS='H' (HLML=2):
{df_habitat_only_hlml2.shape}")


df_habitat_only_hlml2['IRIS'] = df_habitat_only_hlml2['IRIS'].astype('Int64')


gdf_iris['CODE_IRIS_int'] = pd.to_numeric(gdf_iris['CODE_IRIS'], errors='coerce').astype('Int64')


iris_name_map = gdf_iris.set_index('CODE_IRIS_int')['NOM_IRIS']


df_habitat_only_hlml2['Libellé de l\'iris'] = df_habitat_only_hlml2['IRIS'].map(iris_name_map)


geometry_map = gdf_iris.set_index('CODE_IRIS_int').geometry


df_habitat_only_hlml2['geometry'] = df_habitat_only_hlml2['IRIS'].map(geometry_map)


missing_geometries_count = df_habitat_only_hlml2['geometry'].isnull().sum()
if missing_geometries_count > 0:
    print(f"Attention : {missing_geometries_count} IRIS n'ont pas de géométrie correspondante après
le mappage. Leur superficie sera NaN.")


df_habitat_only_hlml2 = gpd.GeoDataFrame(df_habitat_only_hlml2, geometry='geometry',
crs=gdf_iris.crs)


if df_habitat_only_hlml2.crs is None or df_habitat_only_hlml2.crs.is_geographic:
    print(f"CRS actuel est {df_habitat_only_hlml2.crs}. Reprojeter vers EPSG:2154 pour un calcul
précis de la superficie.")
    df_habitat_only_hlml2 = df_habitat_only_hlml2.to_crs(epsg=2154)


df_habitat_only_hlml2['Superficie (km²)'] = df_habitat_only_hlml2.geometry.area / 1_000_000


final_all_columns_in_order = [
    'IRIS',
    'CATIRIS',
    'Libellé de l\'iris',
    'Superficie (km²)',
    'Cadres et professions intellectuelles supérieures',
    'Employés/Employées',
    'Ouvriers/Ouvrières',
    'geometry'
]


df_habitat_only_hlml2 = df_habitat_only_hlml2.loc[:,
~df_habitat_only_hlml2.columns.duplicated(keep='first')]


columns_to_select_and_order = [col for col in final_all_columns_in_order if col in
df_habitat_only_hlml2.columns]
df_habitat_only_hlml2 = df_habitat_only_hlml2[columns_to_select_and_order]


print("DataFrame final avec les colonnes 'Libellé de l'iris' et 'Superficie (km²)' (HLML=2):")
display(df_habitat_only_hlml2.drop(columns=['geometry']).head())
print(f"Dimensions finales du DataFrame enrichi (HLML=2): {df_habitat_only_hlml2.shape}")


output_excel_path_hlml2 = os.path.join(folder_path, 'Brest_données_agrégées_enrichies_non_HLM.xlsx')
output_geojson_path_hlml2 = os.path.join(folder_path,
'Brest_données_agrégées_enrichies_non_HLM.geojson')
```

```
try:
    df_habitat_only_hlml2.drop(columns=['geometry'],
errors='ignore').to_excel(output_excel_path_hlml2, index=False)
    print(f"DataFrame HLML=2 exporté avec succès en Excel: {output_excel_path_hlml2}")
except Exception as e:
    print(f"Erreur lors de l'exportation en Excel pour HLML=2: {e}")


try:
    df_habitat_only_hlml2.to_file(output_geojson_path_hlml2, driver='GeoJSON')
    print(f"GeoDataFrame HLML=2 exporté avec succès en GeoJSON: {output_geojson_path_hlml2}")
except Exception as e:
    print(f"Erreur lors de l'exportation en GeoJSON pour HLML=2: {e}")
```

Appendix 2. Python code for the segregation analysis (D, xPx and xPy indices, location quotient, Moran's I, LISA)

```
!pip install geopandas segregation libpysal esda splot matplotlib-scalebar

from google.colab import drive
drive.mount('/content/drive')
# data comes from my Drive, it's easier to work    with Colab and Drive

import random
import numpy as np
import geopandas as gpd
import pandas as pd

from segregation.singlegroup import Dissim, Isolation, Interaction

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches  # Import mpatches for custom legend entries
from matplotlib.colors import TwoSlopeNorm
from libpysal.weights import Queen
import esda
from splot.esda import moran_scatterplot, lisa_cluster

from matplotlib_scalebar.scalebar import ScaleBar
from matplotlib.ticker import MultipleLocator
from matplotlib.patches import Rectangle

pd.set_option("display.float_format", lambda x: f"{x:0.4f}")

import os

SEED = 12345
random.seed(SEED)
np.random.seed(SEED)

# Definition of input and output paths
INPUT_PATH = '/content/drive/MyDrive/Brest Metropole/Brest_données_agrégées_enrichies.geojson'
OUTPUT_DIR = '/content/drive/MyDrive/Brest Metropole/Resultats_segregation'

os.makedirs(OUTPUT_DIR, exist_ok=True)

IRIS_FILTERED_PATH = os.path.join(os.path.dirname(INPUT_PATH), 'Iris_Brest_filtré.shp')
#for extracting IRIS labels
LIM_ZONE_ETUDE_PATH = os.path.join(os.path.dirname(INPUT_PATH), 'LIM_ZONE_ETUDE.shp') #Neighborhoods
and municipalities of Brest Metropole

LQ_MAP_TITLE_BASE = 'LQ - '
MORAN_SCATTERPLOT_TITLE_BASE = "Moran's Scatterplot - "
LISA_MAP_TITLE_BASE = 'LISA - '

def add_north_arrow(ax, loc='upper right', size_fraction=0.1, color='black'):
    x_pos, y_pos = 0.95, 0.95
    if 'left' in loc:
        x_pos = 0.05

    N_TEXT_HEIGHT_RATIO = 0.3
    ARROW_HEIGHT_RATIO = 0.7
    TEXT_ARROW_GAP_RATIO = 0.1

    # Calculate absolute heights in axes fraction
    total_height_axes_frac = size_fraction
    N_height_axes_frac = total_height_axes_frac * N_TEXT_HEIGHT_RATIO
    arrow_height_axes_frac = total_height_axes_frac * ARROW_HEIGHT_RATIO

    # Adjust vertical positions based on 'loc'
    if 'upper' in loc:
        y_N_center = y_pos - (N_height_axes_frac / 2)
        y_arrow_tip = y_pos - N_height_axes_frac - (arrow_height_axes_frac * TEXT_ARROW_GAP_RATIO)
        y_arrow_base = y_arrow_tip - arrow_height_axes_frac
    else:  # 'lower' in loc
        y_arrow_base = y_pos
        y_arrow_tip = y_pos + arrow_height_axes_frac
```

```
        y_N_center = y_arrow_tip + (N_height_axes_frac / 2) + (arrow_height_axes_frac *
TEXT_ARROW_GAP_RATIO)

    ax.text(x_pos, y_N_center, 'N',
            ha='center', va='center', fontsize=12, color=color,
            transform=ax.transAxes, zorder=10)

    ax.annotate('', xy=(x_pos, y_arrow_tip), xytext=(x_pos, y_arrow_base),
                arrowprops=dict(facecolor=color, edgecolor=color, arrowstyle='simple', lw=1.5,
mutation_scale=20),
                xycoords='axes fraction', textcoords='axes fraction', zorder=9)

print("Loading and preparing the data...")

gdf = gpd.read_file(INPUT_PATH)
gdf['IRIS'] = gdf['IRIS'].astype(str)

GROUPS = {
    'Managers and professionals': 'Cadres et professions intellectuelles supérieures',
    'Employees':                  'Employés/Employées',
    'Workers':                    'Ouvriers/Ouvrières',
}

# Reference population : sum of the three socio-professionnal categories
gdf['Total_3CSP'] = gdf[list(GROUPS.values())].sum(axis=1)

gdf.to_file(os.path.join(OUTPUT_DIR, 'gdf_prepared.geojson'), driver='GeoJSON')

gdf_iris_filtered = gpd.read_file(IRIS_FILTERED_PATH)
gdf_lim_zone_etude = gpd.read_file(LIM_ZONE_ETUDE_PATH)

gdf_lim_zone_etude_wgs84 = gdf_lim_zone_etude.to_crs(epsg=4326)
gdf_iris_filtered_wgs84 = gdf_iris_filtered.to_crs(epsg=4326)

print("Data loaded and prepared.")

print("\nComputing segregation indices...")
total_var = 'Total_3CSP'

# Evenness: dissimilarity index D (group vs rest of the reference population)
# Exposition : indice d'isolement xPx
evenness_results   = {}
exposition_results = {}

for label, col in GROUPS.items():
    evenness_results[label] = {
        'Dissimilarity index (D, group vs rest)': Dissim(gdf, col, total_var).statistic,
    }
    exposition_results[label] = {
        "Isolation index (xPx)": Isolation(gdf, col, total_var).statistic,
    }

df_evenness   = pd.DataFrame(evenness_results)
df_exposition = pd.DataFrame(exposition_results)

print("\n=== EVENNESS DIMENSION ===")
print(df_evenness.round(4))
print("\n=== DIMENSION EXPOSITION ===")
print(df_exposition.round(4))

# Inter-group indices: dissimilarity D(x,y) and interaction xPy
labels = list(GROUPS.keys())

multi_dissim   = pd.DataFrame(index=labels, columns=labels, dtype=float)
multi_interact = pd.DataFrame(index=labels, columns=labels, dtype=float)

for lx in labels:
    for ly in labels:
        if lx == ly:
            continue
        cx, cy = GROUPS[lx], GROUPS[ly]
        pair_total_col = f"__pair_total_{lx}_{ly}"
        gdf[pair_total_col] = gdf[cx] + gdf[cy]

        multi_dissim.loc[lx, ly]   = Dissim(gdf, cx, pair_total_col).statistic
        multi_interact.loc[lx, ly] = Interaction(gdf, cx, pair_total_col).statistic
```

```
        gdf.drop(columns=[pair_total_col], inplace=True)

print("\n=== Dissimilarity index D(x,y) [symmetric] ===")
print(multi_dissim.round(4))
print("\n=== Interaction index xPy [asymmetric: row=x exposed to column=y] ===")
print(multi_interact.round(4))


print("\nSpatialisation of segregation...")


# Location quotient (LQ)
for label, col in GROUPS.items():
    pi_i = gdf[col] / gdf['Total_3CSP']
    P    = gdf[col].sum() / gdf['Total_3CSP'].sum()
    gdf[f'LQ_{label}'] = pi_i / P


gdf_wgs84 = gdf.to_crs(epsg=4326)


df_lq = gdf[['IRIS', "Libellé de l'iris"] + [f'LQ_{l}' for l in GROUPS]].copy()
print("\n=== LOCATION QUOTIENT (LQ) — preview ===")
print(df_lq.round(3).head(10))


# Discretisation
# Fixed references markers :
LQ_BREAKS = [0.5, 0.9, 1.1, 1.5]
LQ_QUALIFIERS = ['very low', 'low', 'neutral', 'high', 'very high']
LQ_COLOR_MAP = dict(zip(LQ_QUALIFIERS, ['#2166ac', '#92c5de', '#f7f7f7', '#f4a582', '#b2182b']))
NO_DATA_COLOR = 'lightgrey'
MASTER_EDGES = [-np.inf] + LQ_BREAKS + [np.inf]  # is used to determine the correct descriptor for
each interval


fig, axes = plt.subplots(1, len(GROUPS), figsize=(9 * len(GROUPS), 9))
for ax, label in zip(axes, GROUPS):
    lq_col = f'LQ_{label}'
    lq_min = gdf[lq_col].min()
    lq_max = gdf[lq_col].max()



    inner_breaks = [b for b in LQ_BREAKS if lq_min < b < lq_max]
    bin_edges = [lq_min] + inner_breaks + [lq_max]


    bin_labels = []
    for i in range(len(bin_edges) - 1):
        midpoint = (bin_edges[i] + bin_edges[i + 1]) / 2
        idx = min(max(np.searchsorted(MASTER_EDGES, midpoint) - 1, 0), len(LQ_QUALIFIERS) - 1)
        bin_labels.append(LQ_QUALIFIERS[idx])


    class_col = f'{lq_col}_classe'
    gdf[class_col] = pd.cut(gdf[lq_col], bins=bin_edges, labels=bin_labels, include_lowest=True,
duplicates='drop')


    gdf_iris_filtered_wgs84.plot(ax=ax, facecolor=NO_DATA_COLOR, edgecolor='grey', linewidth=0.3)


    missing = gdf_wgs84[gdf[class_col].isna()]
    if not missing.empty:
        missing.plot(ax=ax, color=NO_DATA_COLOR, edgecolor='grey', linewidth=0.3)


    for qualifier in bin_labels:
        subset = gdf_wgs84[gdf[class_col] == qualifier]
        if not subset.empty:
            subset.plot(ax=ax, color=LQ_COLOR_MAP[qualifier], edgecolor='grey', linewidth=0.3)


    gdf_lim_zone_etude_wgs84.plot(ax=ax, facecolor='none', edgecolor='black', linewidth=1.0)
    ax.set_title(f'{LQ_MAP_TITLE_BASE}{label}')
    ax.set_xlabel('Longitude')
    ax.set_ylabel('Latitude')
    ax.set_aspect(1.3)
    add_north_arrow(ax)
    center_lat = (ax.get_ylim()[0] + ax.get_ylim()[1]) / 2
    km_per_degree_at_center_lat = 111.320 * np.cos(np.deg2rad(center_lat))
    meters_per_degree_at_center_lat = km_per_degree_at_center_lat * 1000
    scalebar = ScaleBar(dx=meters_per_degree_at_center_lat, units="m", location="lower left",
rotation='horizontal-only')
    ax.add_artist(scalebar)


    legend_handles = [mpatches.Patch(facecolor=NO_DATA_COLOR, edgecolor='grey', label='No data')]
    for i, qualifier in enumerate(bin_labels):
        range_label = f"{bin_edges[i]:.2f} - {bin_edges[i + 1]:.2f}   {qualifier}"
        legend_handles.append(mpatches.Patch(facecolor=LQ_COLOR_MAP[qualifier], edgecolor='grey',
label=range_label))
```

```
    legend = ax.legend(
        handles=legend_handles, loc='upper left', title='Location Quotient',
        fontsize=8, title_fontsize=9, frameon=True, framealpha=0.9
    )
    legend.get_title().set_fontweight('bold')
    ax.add_artist(legend)
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'carte_LQ.png'), dpi=150)
plt.show()


# Queen contiguity
w = Queen.from_dataframe(gdf, use_index=False)
w.transform = 'r'
print(f"\nWeights matrix: {w.n} units, {len(w.islands)} isolated island(s), "
      f"{np.mean(list(w.cardinalities.values())):.2f} neighbours on average")


# Moran index
N_PERMUTATIONS = 999
np.random.seed(SEED)


moran_summary = {}
moran_objects = {}
for label in GROUPS:
    m = esda.Moran(gdf[f'LQ_{label}'].values, w, permutations=N_PERMUTATIONS)
    moran_objects[label] = m
    moran_summary[label] = {
        "Moran's I Index":            m.I,
        "Expected value under H0 (E[I])": m.EI,
        "z-score (permutations)":         m.z_sim,
        "p-value (permutations)":         m.p_sim,
        "Significant at 5 %":             "Yes" if m.p_sim < 0.05 else "No",
    }
df_moran = pd.DataFrame(moran_summary)
print("\n=== GLOBAL MORAN'S INDEX (on Location Quotient) ===")
print(df_moran.round(4))


# Moran's scatterplots
fig, axes = plt.subplots(1, len(GROUPS), figsize=(6 * len(GROUPS), 5.5))
for ax, label in zip(axes, GROUPS):
    moran_scatterplot(moran_objects[label], ax=ax, aspect_equal=False)
    ax.set_title(f"{MORAN_SCATTERPLOT_TITLE_BASE}{label}\nI = {moran_objects[label].I:.3f} "
                 f"(p = {moran_objects[label].p_sim:.3f})")
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'moran_scatterplots.png'), dpi=150)
plt.show()


# LISA
ALPHA = 0.05
lisa_objects = {}
df_lisa = gdf[['IRIS', "Libellé de l'iris"]].copy()
for label in GROUPS:
    lisa = esda.Moran_Local(gdf[f'LQ_{label}'].values, w, permutations=N_PERMUTATIONS, seed=SEED)
    lisa_objects[label] = lisa
    df_lisa[f'LISA_{label}_cluster'] = np.array(lisa.get_cluster_labels(crit_value=ALPHA))
    df_lisa[f'LISA_{label}_I_local'] = lisa.Is
    df_lisa[f'LISA_{label}_p_sim']   = lisa.p_sim


print("\n=== DISTRIBUTION OF LISA CLUSTERS BY GROUP ===")
for label in GROUPS:
    print(f"\n{label} :")
    print(df_lisa[f'LISA_{label}_cluster'].value_counts())


# LISA clustering
fig, axes = plt.subplots(1, len(GROUPS), figsize=(6 * len(GROUPS), 6))
for ax, label in zip(axes, GROUPS):
    lisa_cluster(lisa_objects[label], gdf_wgs84, p=ALPHA, ax=ax)
    gdf_lim_zone_etude_wgs84.plot(ax=ax, facecolor='none', edgecolor='black', linewidth=1.0)
    ax.set_title(f'{LISA_MAP_TITLE_BASE}{label}')
    ax.set_xlabel('Longitude')
    ax.set_ylabel('Latitude')
    ax.set_aspect(1.3)
    ax.tick_params(axis='both', which='major', labelsize=8)
    ax.xaxis.set_major_locator(MultipleLocator(0.05))
    ax.yaxis.set_major_locator(MultipleLocator(0.025))
    xmin, xmax = ax.get_xlim()
    ymin, ymax = ax.get_ylim()
    rect = Rectangle((xmin, ymin), xmax - xmin, ymax - ymin,
                     fill=False, edgecolor='black', linewidth=1.5, zorder=100)
    ax.add_patch(rect)
    ax.grid(True, which='major', linestyle=':', color='grey', alpha=0.5)
    add_north_arrow(ax)
    center_lat = (ax.get_ylim()[0] + ax.get_ylim()[1]) / 2
    km_per_degree_at_center_lat = 111.320 * np.cos(np.deg2rad(center_lat))
```

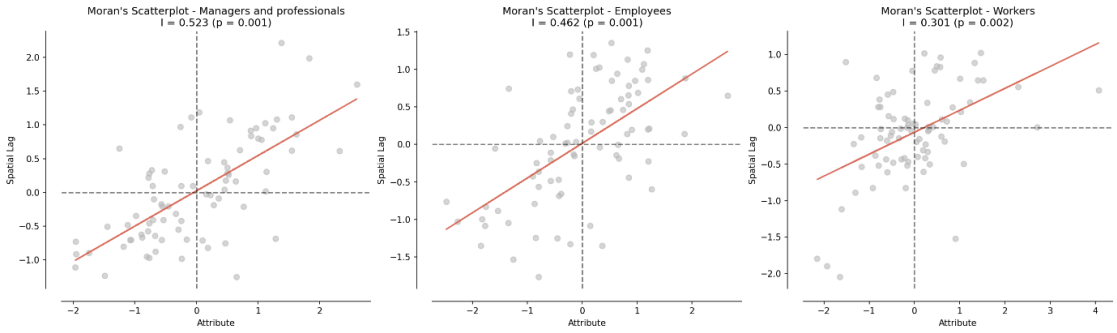
```
    meters_per_degree_at_center_lat = km_per_degree_at_center_lat * 1000
    scalebar = ScaleBar(dx=meters_per_degree_at_center_lat, units="m", location="lower left",
rotation='horizontal-only')
    ax.add_artist(scalebar)
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'carte_LISA.png'), dpi=150)
plt.show()


# Exporting results to an Excel file
print("\nExporting all tabular results to Excel files...")

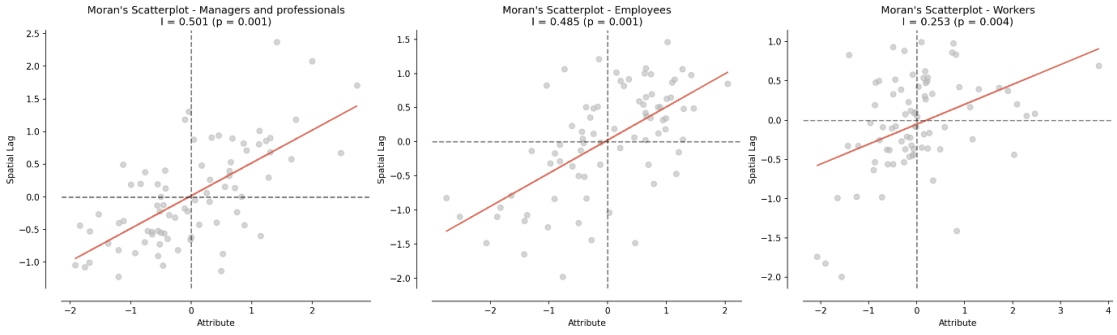

with pd.ExcelWriter(os.path.join(OUTPUT_DIR, 'indices_segregation.xlsx')) as writer:
    df_evenness.to_excel(writer, sheet_name='Dissimilarite groupe-reste')
    df_exposition.to_excel(writer, sheet_name='Isolement')
    multi_dissim.to_excel(writer, sheet_name='Dissimilarite D(x,y)')
    multi_interact.to_excel(writer, sheet_name='Interaction')
print(f"Segregation indices exported to: {os.path.join(OUTPUT_DIR, 'indices_segregation.xlsx')}")


with pd.ExcelWriter(os.path.join(OUTPUT_DIR, 'geospatial_results.xlsx')) as writer:
    df_lq.to_excel(writer, sheet_name='Quotients Localisation', index=False)
    df_moran.to_excel(writer, sheet_name='Moran Global')
    df_lisa_summary = gdf[['IRIS', "Libellé de l'iris"]].copy()
    for label in GROUPS:
        df_lisa_summary[f'LISA_{label}_cluster'] = df_lisa[f'LISA_{label}_cluster']
        df_lisa_summary[f'LISA_{label}_I_local'] = df_lisa[f'LISA_{label}_I_local']
        df_lisa_summary[f'LISA_{label}_p_sim'] = df_lisa[f'LISA_{label}_p_sim']
    df_lisa_summary.to_excel(writer, sheet_name='LISA Clusters', index=False)
print(f"Geospatial results exported to: {os.path.join(OUTPUT_DIR, 'geospatial_results.xlsx')}")
print(f"\nPipeline complete, all results (indices and maps) were saved in: {OUTPUT_DIR}")
```

Appendix 3. Moran scatterplots, excluding social housing



Appendix 4. Moran scatterplots, including social housing



Appendix 5. Overview of all the maps

For better legibility, all the maps can be viewed on an interactive page at the following address: https://mo-soidridine.github.io/brest_metropole/.